

DERIVATIVES 101 • DECK 0

Foundations

The handful of building blocks that every product on this desk is made from.
No prior finance knowledge assumed.

Six decks, one idea repeated

0

Foundations

Spot, options, premium, volatility, barriers, coupons, autocalls, baskets. The vocabulary everything else reuses.

1

KODA / KODD

Accumulators and decumulators. Buy a little every day at a discount — until the market moves against you.

2

Reverse convertibles

Equity and FX. A high coupon against your principal. Conditional coupons, callable, autocallable, lock-in.

3

Asian FCN & ELN

The short-dated, worst-of, physically-settled notes that dominate Hong Kong, Singapore and Taiwan flow.

4

Discount & bonus

The European listed side. Premium taken as a price discount or a conditional bonus instead of a coupon.

5

Protection & leverage

Capital protected notes at one end, mini futures at the other. Same issuer credit underneath both.

Every product in decks 1–5 is the same trade wearing a different suit: the client sells optionality and gets paid for it.

Four investment families, plus leverage

11xx	12xx	13xx	14xx	2xxx
Capital protection	Yield enhancement	Participation	Additional credit risk	Leverage
The floor is bought first; whatever is left buys upside.	Capped upside in exchange for a known return. The biggest family by far.	Open-ended upside. Trackers, outperformance, bonus certificates.	A second credit exposure bolted on top of the issuer's own.	Warrants, knock-outs, mini futures, constant leverage.
Deck 5	Decks 1-4	Deck 4	Not covered	Deck 5

The family is decided by the shape of the return, not by how the product is built. Add a cap to a bonus certificate and it moves from participation to yield enhancement, even though only one option leg changed.

Two traps: the Swiss map has no code for a capped bonus, and 1260 means different products in Switzerland and the rest of Europe.

Structured products are recipes, not ingredients

- A structured product is a bundle. Unwrap it and you find a deposit or bond, plus one or more options.
- The client is almost always a net seller of options — that is where the headline yield or discount comes from.
- The bank is the buyer of those options, and then spends the product's life hedging what it bought.
- So there are only two questions worth asking about any new payoff: what has the client sold, and what makes it worse?

A yield-enhancement note =

Zero-coupon bond

Repays 100 at maturity. Costs less than 100 today — the gap is your budget.

+ option premium received

The client sells a put (or a strip of them). The premium buys the coupon.

– issuer margin & hedging cost

What the desk keeps, and what it expects to pay to stay hedged.

= the coupon on the term sheet

Everything the client sees is the residual of the three lines above.

Spot, forward, and why the forward is not a forecast

Spot

The price you pay to own the asset right now. HSBC at HK\$95.40. USD/JPY at 147.20.

Forward

The price agreed today for delivery on a future date. Fixed now, settled later.

Carry

Forward = spot adjusted for interest earned, funding paid, and dividends foregone. Nothing else.

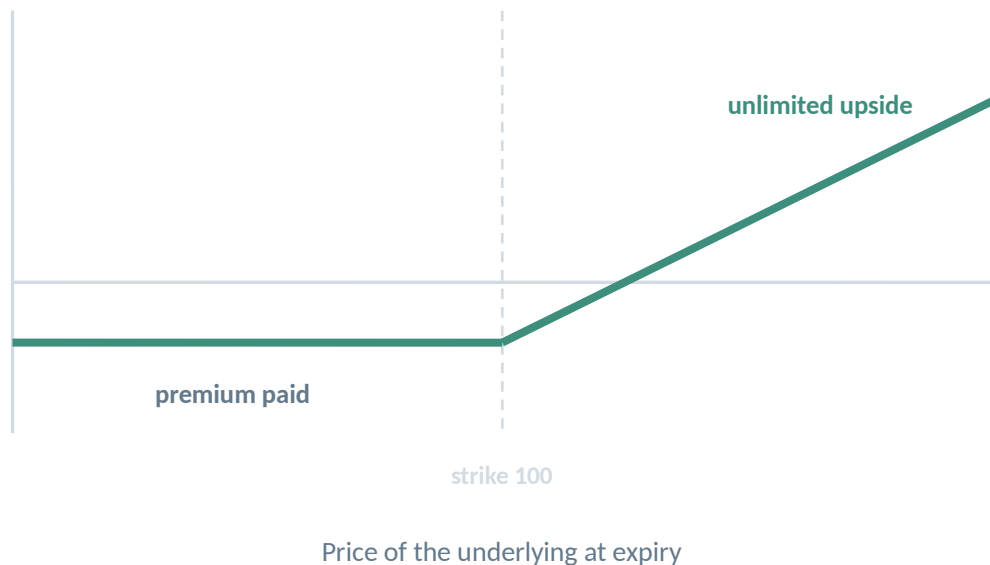
A one-year forward on a stock trades below spot when the dividend yield exceeds the funding rate. That is arithmetic, not a view.

This matters because clients regularly read a 'discount to spot' on a term sheet as a bargain, when part of it is simply carry. The rest — the part that actually pays for the product — is option premium.

BUILDING BLOCK 2

A right for one side, an obligation for the other

Long call — the right to buy at 100



Short put — the obligation to buy at 100



Almost every product we trade puts the client on the right-hand chart. A small capped gain, paid up front, against a loss that runs all the way to zero.

Premium is the price of uncertainty

- **Volatility** measures how much the underlying moves, not which way.
- Realised volatility is what actually happened. Implied volatility is what the option market is charging for what might happen.
- Higher implied vol means a bigger premium — so a scared market funds a fatter coupon for the same risk barrier.
- That is why coupons on the same structure jumped sharply in April 2025 when the VIX spiked, without a single term on the sheet changing.

Vol ↑ → Premium ↑

More movement priced in, so the option the client sells is worth more.

Tenor ↑ → Premium ↑

More time, more chance of a big move — but also more time to be wrong.

Strike ↓ → Premium ↓

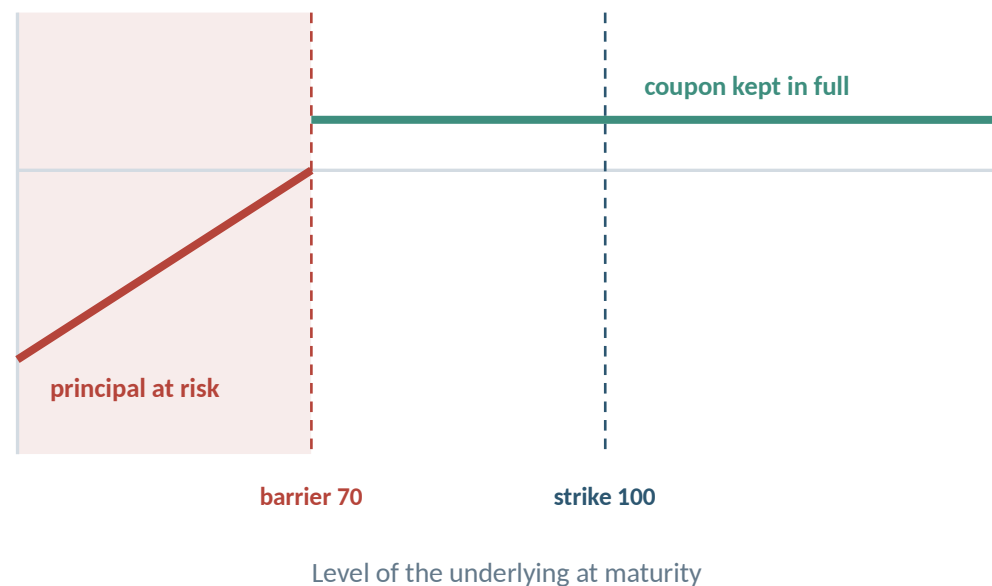
A put struck further below spot is less likely to be exercised, so it pays less.

Assets ↑ → Premium ↑

A worst-of basket sells more optionality, so the coupon rises with each name added.

Two charts tell you almost everything

Payoff diagram — what you get, given where it ends



Path chart — whether the route mattered



If the product has a barrier, the payoff diagram alone is a lie. You have to know whether the barrier was touched.

Barriers: the same product, two different lives

Knock-out (KO)

Above spot, usually. Touch it and the trade ends early — often profitably for the client, but the profit is capped and the trade is gone.

Also called the autocall or early-redemption trigger.

Knock-in (KI)

Below spot. Touch it and the client's downside switches on. Until then the note behaves like a bond; afterwards it behaves like the stock.

Hong Kong retail ELIs call the protection above this level an 'airbag'.

Observation style changes the price more than most people expect

European

Observed once, at maturity. Only the final close matters.

American / continuous

Observed every tick of every day. The low of the year is what counts.

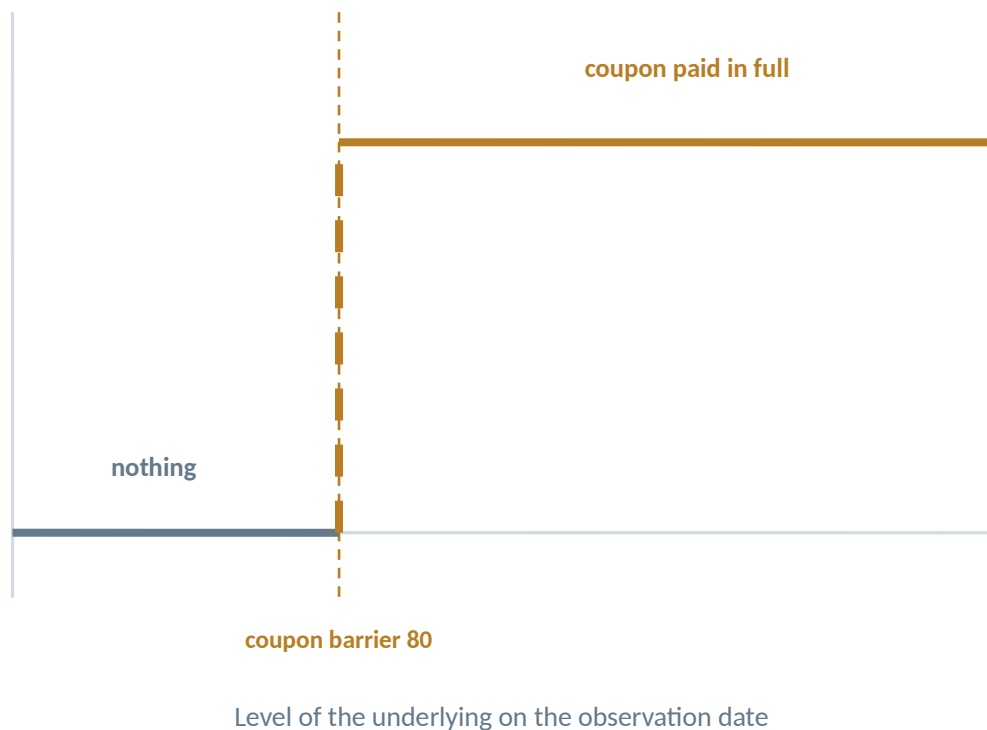
Discrete / daily close

Observed on scheduled dates only — daily, monthly or at each coupon date.

The market has drifted from continuous towards European knock-ins since 2015 — a quiet but material de-risking of the whole product set.

Coupons: fixed, conditional, or remembered

A conditional coupon is a digital option



Fixed / unconditional

Paid whatever the underlying does. The defining feature of a Fixed Coupon Note.

Conditional

Paid only if the underlying is above a coupon barrier on the observation date.

Memory

Missed coupons are not lost — they are banked and paid later if the barrier is regained.

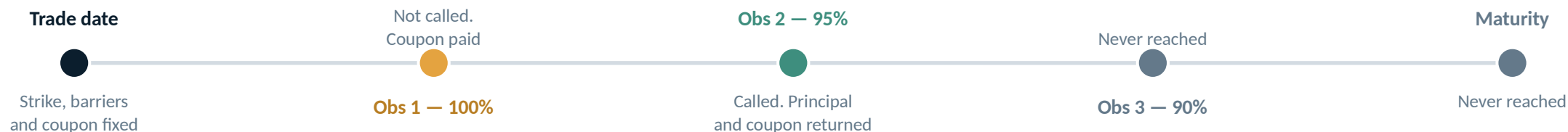
Accrual

Accrues only on the days the underlying closes above a level. Part-paid coupons are normal.

A coupon that is 'guaranteed' is guaranteed in isolation. It is never a statement about the principal.

Autocall: the product decides when it ends, not you

A three-year note that usually ends long before three years



Step-down triggers

Each observation calls at or below the previous level, so the note gets easier to redeem as it ages. 85 / 85 / 85 / 80 / 75 / 70 is a normal Korean schedule.

Non-call period

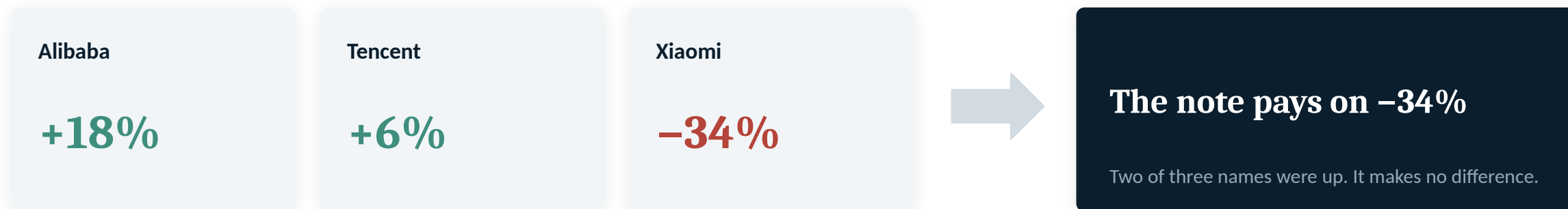
Three to six months at the start where no call can happen. It guarantees the client at least one or two coupons.

Why it matters to us

Expected life is not contractual life. Cashflow schedules, accruals and position ageing all have to handle early termination as the normal case, not the exception.

Autocall probability is why a client who wants three years of coupons rarely gets them: a rising market redeems the note early.

Worst-of: the basket that only remembers its weakest name



- More names means a higher coupon — and a strictly worse expected outcome.
- Adding a fourth name cannot improve the worst performer. It can only make it worse. The extra coupon is the price of that.
- The extra coupon is paid for with correlation. A worst-of option is worth most when the names decouple, because the worst performer is then further from the pack — so the client is selling dispersion without ever being told so.
- Two to four names is the market norm. Three is the most common structure we see.

Three levers, and you cannot pull all three

1 Coupon

How much the client is paid

Raise it and something else must give.

2 Protection

How far the market can fall before it hurts

A deeper barrier costs coupon.

3 Participation

How much upside the client keeps

Most of these products keep none.

When a client asks for a higher coupon, they are asking for a worse barrier, a shorter fuse, or another name in the basket. There is no free version.

This is the single most useful sentence to have in your head when reading a term sheet. Every unusual number on the page is paid for by another number on the same page.

The desk buys the option, then spends a year defending it

We are long the downside option

The client sold it to us. On day one that is a comfortable place to be.

We are short the coupon

Paid whether or not the hedge performs. It is a fixed liability against a moving asset.

The hedge is not static

As spot falls, our sensitivity to volatility rises, peaks, and then collapses — forcing us to buy volatility back at the worst possible moment.

Everyone flips at once

The whole street issues on the same names at similar levels. When the hedge turns, it turns for all of us simultaneously, in the same direction.

Deck 2 covers this properly. For now: the client's risk and the desk's risk are not mirror images — they fail in different ways, at different times.

The anatomy of a termsheet

1 Classification line

The SSPA / EUSIPA category and code, plus the settlement and headline feature. Read this before anything else — it tells you which family you are in.

2 Distribution restriction

Who may be sold this, under which regime. In Switzerland this is the FinSA line, and it is the reason the same payoff comes in different wrappers.

3 Terms

Issue price, denomination, issue size, coupon, maximum yield, tenor, and the four dates. Everything commercial is in this block.

4 Underlying

Initial Level, Strike, Barrier, Trigger Barrier, Payout Threshold, Ratio, Valuation Time. Every level is quoted twice — absolute and as a percentage of initial.

5 Redemption

The payoff, written as nested conditions. Final Redemption Amount, then Barrier Event, then the observation dates and period.

6 Early redemption & payout

Trigger Redemption for the autocall, Payout for the coupon. Each has its own barrier, its own observation dates and its own table.

Risk factors come last in the document and matter first. Read backwards on your second pass.

Nine terms that decide the whole payoff

Initial Level	The reference fixed on the Initial Fixing Date. Everything else is quoted as a percentage of it.
Strike	The level the payoff is measured against, and the price at which shares are delivered.
Barrier	Where principal protection is lost. Check whether it is observed continuously or only at the final fixing.
Trigger Barrier	The autocall level. A Trigger Event redeems the note early at the Trigger Redemption Price.
Payout Threshold	The level a conditional coupon needs. It is a separate number from the Barrier and can sit either side of it.
Ratio	$\text{Denomination} \div \text{Strike}$. The number of shares delivered on physical settlement, rounded down, with cash for the fraction.
Maximum Yield	The most the product can pay. Compare it with the headline coupon before you believe the headline.
Valuation Time	Usually the scheduled closing time — but not always, and for FX often a named time in a named city.
The four dates	Initial Fixing, Issue, Final Fixing, Final Redemption. They are all different, and the gaps between them are real.

Two weeks can separate the Initial Fixing Date from the Issue Date. The trade is struck long before the money moves.

The words you will hear on day one

Notional	The face amount the payoff is calculated on. Not what the client paid.	Fixing	The scheduled observation of a price. Fixings drive everything.
Strike	The reference level for the option. Often set below spot.	Physical settlement	The client is delivered shares instead of cash.
Knock-in	The level below which the client's downside switches on.	Vega	Sensitivity to implied volatility.
Knock-out / autocall	The level at which the trade terminates early.	Skew	The market charging more for downside options than upside ones.
Worst-of	Payoff references the weakest performer in a basket.	Tenor	Contractual life. Rarely the actual life.
Gearing	A multiplier applied to the loss side. Usually 2x.	PI / AI / SPI	Professional, accredited and sophisticated investor categories.

Five things, and then we start on real products

- 1 Unwrap the bundle. Bond plus options, always.
- 2 Find what the client sold. It is nearly always a put, or a strip of them.
- 3 Check whether the path matters. Barriers make the payoff diagram incomplete.
- 4 Assume early termination. Contractual tenor is the exception, not the rule.
- 5 Remember the trade-off. A better number on the term sheet was paid for by a worse one.

DERIVATIVES 101 • DECK 1

KODA & KODD

Knock-out discount accumulators and decumulators.

The biggest structured product in Hong Kong, and the one that has hurt clients most.

IN ONE SENTENCE

A contract to keep buying, a little every day, at a discount

The client is obliged to buy a fixed quantity of the underlying every trading day, at a price below today's spot, until the price rises through a knock-out barrier.

Two catches. Below the strike, the daily quantity doubles. And the knock-out caps the upside at a few percent, while the downside runs all the way to a share price of zero.

Obligation

Not a right. The client must buy every single day, whatever has happened.

Discount

The strike sits 5–15% below spot. That discount is option premium, not a gift.

Gearing

Below strike the daily size doubles. You buy most where you are most wrong.

Knock-out

A rally ends the trade early, with a small profit. That is the good outcome.

Hong Kong investors renamed it in 2008. 'Accumulator' became 'I kill you later' — a homophone that stuck, and is now used in the regulator's own investor-education material.

Reading an accumulator term sheet

Reference Underlying	A single listed share, an FX pair, or occasionally an index
Reference Spot	USD 320.00 (100%)
Forward Price	USD 288.00 (90%) — the price the client is obliged to buy at
Knock-Out Price	USD 336.00 (105%), on the closing price, from the trade date
Underlyings per Day	10
Leverage	2 — so 20 a day when the close is below the Forward Price
Settlement Frequency	Monthly, in arrears, physical delivery
Number of Observation Periods	12
Guaranteed Observation Period(s)	2 — the first two months accumulate whatever happens
Max. Scheduled Trading Days	250
Max. Total No. of Underlyings	$250 \times 20 = 5,000$
Max. Exposure	$5,000 \times \text{USD } 288.00 = \text{USD } 1,440,000$

Three numbers to check first

Max. Exposure

Not the margin. This is the real size of the obligation — USD 1.44m against roughly USD 432,000 posted.

Forward Price to Knock-Out

288 to 336 is the entire window in which this trade works. Fifteen percent of spot.

Gearing multiple

2x is standard. 3x exists. It changes the maximum loss, not the maximum gain.

THE CORE PICTURE

What one observation day is worth

Client P&L for a single observation day



- The whole trade is this chart, repeated 120 times.
- Above the Forward Price the client makes a little. The most any single day can earn is the gap up to the Knock-Out Price — and reaching it ends the trade.
- Below the Forward Price the slope doubles. There is no barrier and no airbag on this side — the only floor is a share price of zero.
- Note where the chart is truncated on the right and open-ended on the left. That asymmetry is the product.

Ten units bought on any day the close is at or above the Forward Price; twenty on any day it closes below. The Knock-Out is tested on the close, from the trade date onwards.

WORKED EXAMPLE

Same contract, three markets, three very different outcomes

Rally

Knocks out on day 15 — but two periods are guaranteed

Days accumulated 42

Units bought 420 @ 288

Cost USD 120,960

Value at 336 USD 141,120

P&L **+USD 20,160**

Return on margin **+4.7%**

Sideways

Drifts to 300, never knocks out

Days accumulated 250

Units bought 2,500 @ 288

Cost USD 720,000

Value at 300 USD 750,000

P&L **+USD 30,000**

Return on margin **+6.9%**

Fall

Gaps below the Forward Price in week one, slides to 160

Days accumulated 250 (all geared)

Units bought 5,000 @ 288

Cost USD 1,440,000

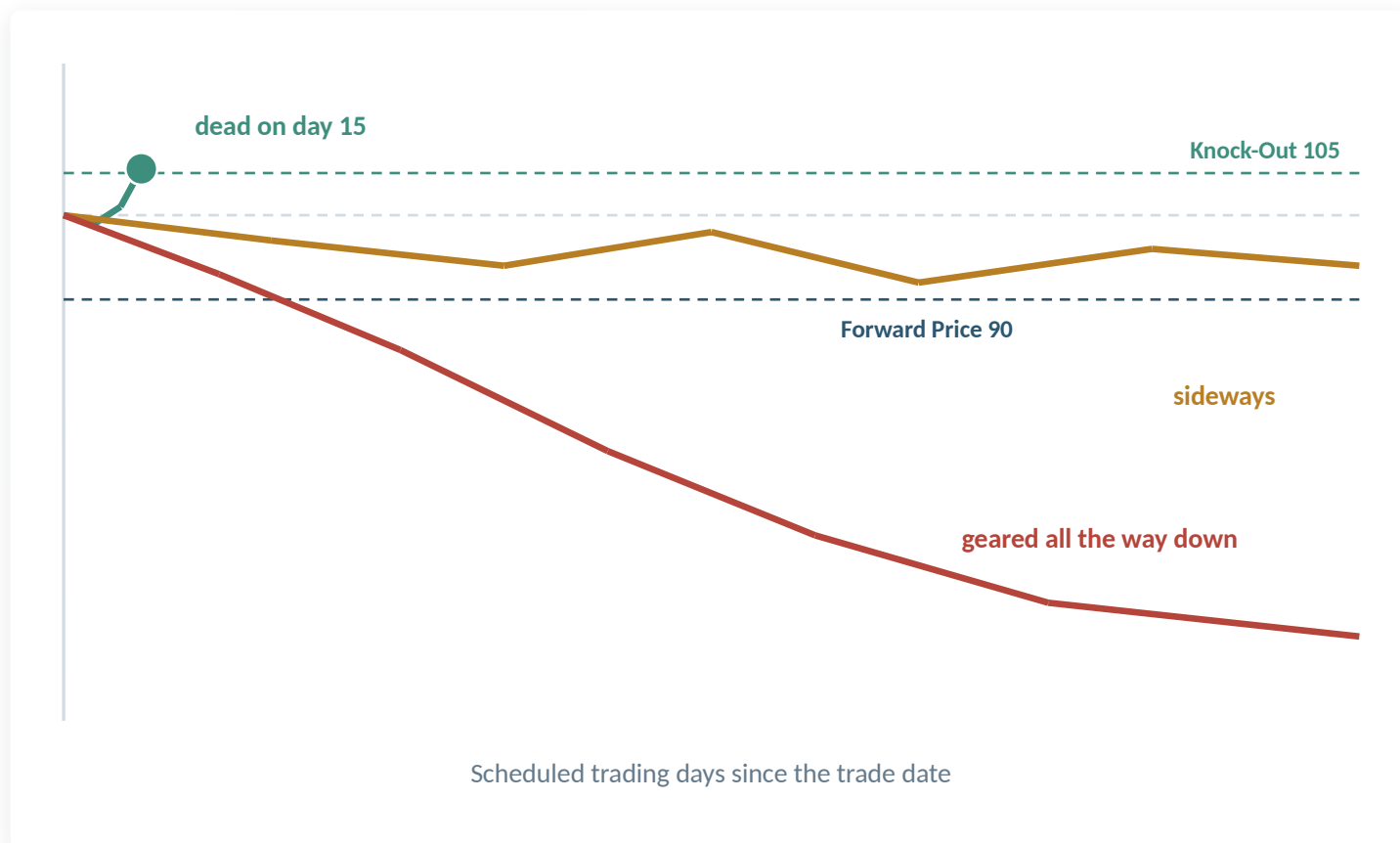
Value at 160 USD 800,000

P&L **-USD 640,000**

Return on margin **-148%**

Best realistic case +6.9%. Bad case -148%. If the share went to zero the client would owe the full USD 1.44m Max. Exposure against roughly USD 432,000 posted — 3.3× the margin.

The knock-out is a trapdoor, not a target



- **The client's good scenarios are short and small.**
- A rally does not make the client rich — it terminates the trade after three weeks with a few percent of margin earned, and the guaranteed periods still run to the end of month two.
- The client's bad scenario is long and large. Nothing stops the accumulation. Every day the stock is lower, the client buys twice as much of it.
- **There is no symmetry here. The trade is designed so that the pleasant outcome ends quickly and the unpleasant one runs to term.**

Decumulators: the mirror image, with no ceiling

Client P&L for a single observation day



The pitch

'Sell your concentrated holding at a premium to today's price.' Attractive to founders and families sitting on one large position.

The catch

If the stock rallies, the client is obliged to sell twice as much, at a price that gets worse every day. A share price has no ceiling.

Why regulators single it out

Accumulator losses are capped at the full notional. Decumulator losses are not capped at all. The SFC and HKMA say so explicitly.

Terms mirror the accumulator: Forward Price 114% of spot, Knock-Out 95%. The client sells 10 units a day below the Forward Price and 20 above it.

A strip of barrier options, priced as one trade

1

Long 1 up-and-out call per day

Struck at the Forward Price, knocked out at 105%. This is the client's small daily gain when the share behaves.

2

Short 2 up-and-out puts per day

Same strike, same knock-out. The barrier sits above spot, so both legs die on a rally. This is where the discount — and the loss — comes from.

3

× 250 scheduled trading days

Each day is its own pair of options with its own expiry. The 'trade' is a portfolio of 750 options.

What that means for risk

Short volatility

Net, the client has sold twice as many options as bought. A vol spike hurts the mark before the stock has moved at all.

Short skew

The puts sold are below spot, where the market charges the most. Skew steepening in a selloff hits the mark again.

Path dependent

Two markets ending at the same price can produce completely different P&L, depending on whether the barrier was touched.

Non-linear in size

Max. Exposure is realised exactly in the loss case. Size grows as the trade goes wrong.

THE QUESTION EVERY CLIENT ASKS

Where does the discount actually come from?

“The share purchase discount price in the stock accumulator contract in fact comes from the premium the investor receives from selling the options to the counterparty.”

Meena Datwani, then Executive Director, Hong Kong Monetary Authority

It is not a discount

It is a payment for risk the client has agreed to take. Priced fairly, its expected value is roughly zero before costs.

It is not dollar-cost averaging

DCA buys a constant amount regardless of price. An accumulator buys double when the price falls and stops entirely when it rises. It is the opposite of DCA.

It is not 'I wanted the stock anyway'

Wanting HK\$3m of a stock is not the same as an HK\$21m obligation to keep buying it. Size, not preference, is what breaks clients.

These three sentences are the entire content of the regulators' concern about how this product is sold.

Most clients are not stopped out by the payoff — they are stopped out by margin

- **The mark-to-market is an option valuation, not intrinsic value.**
- The position can lose value on a rise in implied volatility alone, with the share price unchanged.
- Initial margin is typically 30% of maximum notional for equity accumulators, 10% for FX. Banks reserve the right to raise it — up to 100% — during the life of the trade.
- If the client cannot meet the call, the bank closes out at its own price. There is no exchange, no screen, and no obligation to unwind gradually.
- **This is how most of the 2008 losses were actually realised: forced unwind cost, not accumulated shares.**

Shine Grace Investment Ltd v Citibank

Nine equity accumulators bought over two days in October 2007. Three knocked out within a month. The remaining six were unwound in January 2008.

HK\$427m

Unwind cost

Paid to close the six surviving contracts early

HK\$51m

Loss on shares

The part that came from actually holding the stock

WHO BUYS THESE, AND WHY

The pitch and the payoff, side by side

What the client hears

"Buy the stock you like at a 12% discount."

"It's disciplined phased entry."

"Capital efficient — only 30% down."

"You have a range-bound view."

What they have actually agreed to

You are paid a premium to accept a leveraged obligation to keep buying it while it falls.

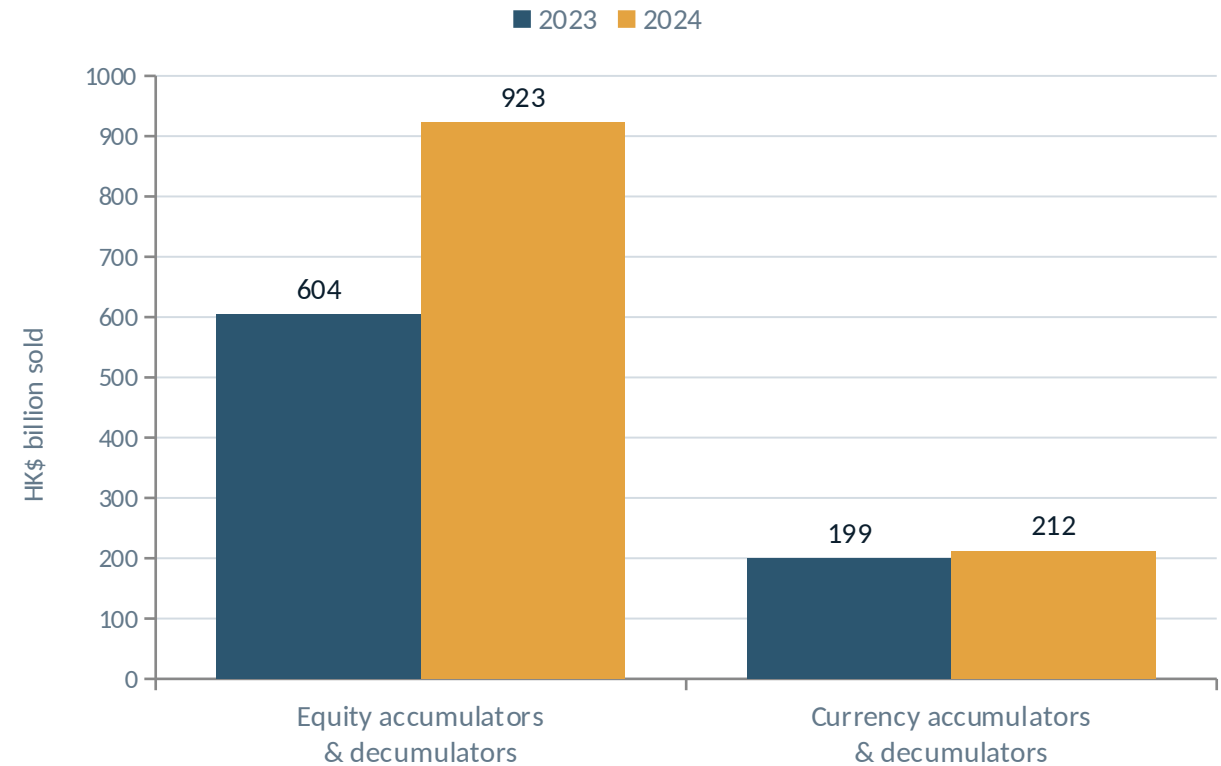
Phased entry stops when you want it to. This stops only when you are winning.

Leverage of 3.3×. The margin is the smaller number on the page, not the risk.

Correct — and this is one of the only instruments that monetises that view. It is not a bad product, it is a misunderstood one.

The last row is the important one. Accumulators are a legitimate way to express a sideways view. Almost nobody who loses money on one had that view.

This is not a niche product in Asia



44%

of HK structured products

Accumulators and decumulators
combined, 2024

1.2m

clients, all product types

Buyers of non-exchange-traded products,
all types; 90% not PIs

≈ HK\$1.1 trillion

of accumulators and decumulators sold in Hong Kong in 2024
alone — against roughly US\$23bn outstanding at the 2008 peak.

Source: SFC–HKMA Joint Product Survey 2023 and 2024. Figures are annual sales notional, not outstanding positions.

IT HAS GONE WRONG BEFORE

Three episodes worth knowing by name

2008

Hong Kong

Roughly US\$23bn of accumulators outstanding by April 2008; far more once gearing is counted. Individual private-bank losses ran into the hundreds of millions of HK dollars. The product acquired its nickname here.

2014–17

Taiwan TRFs

Leveraged CNH target-redemption forwards — FX accumulators in all but name, sold to exporters as hedging. After the August 2015 renminbi devaluation, Taiwanese SMEs were estimated to have lost around US\$6.6bn.

Jan 2024

China snowballs

A different product with the same DNA. Around US\$50bn notional outstanding with roughly 40% of knock-in levels already breached; hedge unwinds pushed CSI 1000 futures limit-down on 22 January.

The common thread is not the asset class. It is leverage, a capped upside, and a client base that read 'discount' as 'safe'.

Clients lose in court. Banks lose to regulators.

- **Hong Kong courts have consistently found for the banks.**
- Shine Grace v Citibank (HK Court of Appeal, 2022): a HK\$478m accumulator loss, claim dismissed. Volunteering advice did not create a duty to advise on suitability where the contract said otherwise.
- Deutsche Bank v Chang Tse Wen (Singapore Court of Appeal, 2013): a US\$49m first-instance award to the client, reversed in full on appeal.
- **The exposure is regulatory and remediation-driven, not litigation-driven.**
- HSBC Private Bank (Suisse) was fined HK\$400m in 2017 — then a Hong Kong record — over 2003–08 sales that included forward accumulators.

What changed recently

HKMA, 14 March 2025

A single consolidated Guidance on Selling of Accumulators replaced seven legacy circulars. No new obligations, but it is now the one document to read.

SFC / HKMA, July 2023

The Sophisticated Professional Investor regime — HK\$40m portfolio or HK\$80m net assets plus a knowledge test — streamlines the transaction-level suitability requirements. Much accumulator flow now sits under it.

Joint thematic review, 2024

Found firms failing to capture leverage and volatility in product due diligence, and recorded a salesperson unable to explain accumulator mechanics beyond reading the term sheet aloud.

Five things about accumulators

- 1 It is an obligation to buy, not an option to buy. Every trading day, whatever happens.
- 2 The discount is option premium the client has sold. It is not a bargain and it is not carry.
- 3 Gearing means the client buys most of the position in exactly the scenario they did not want.
- 4 Maximum notional, not margin, is the size of the trade. Typically 3–4× the cash posted.
- 5 The upside ends early and small; the downside runs to term and large. That asymmetry is the entire product.

DERIVATIVES 101 • DECK 2

Reverse convertibles

Equity and FX. Plus the callable, autocallable and lock-in variants that make up most of what actually trades.

IN ONE SENTENCE

A high coupon, paid for with your principal

A note that pays an above-market coupon, and repays your money in full only if the underlying stays above a level. Below it, you get the asset instead of your cash.

It is 'reverse' because a normal convertible lets the investor choose to convert. Here the issuer decides — and only ever converts when it hurts.

Deposit or bond

Repays 100 at maturity. This is the wrapper.

+ short put

Sold by the client, struck at or below spot.
This is the risk.

= high coupon

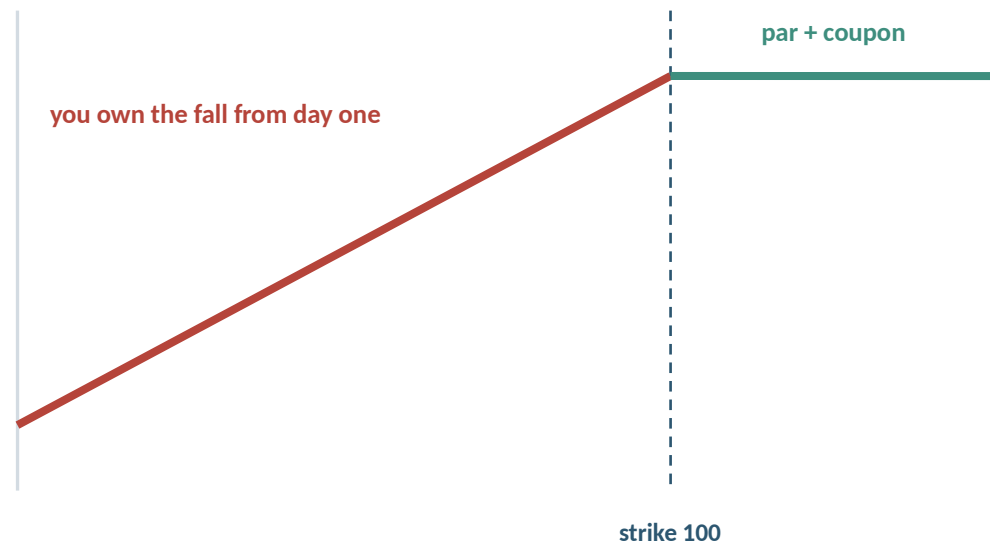
The put premium, spread over the life of the note.

If you understood the short put in Deck 0, you already understand this product. Everything from here is packaging.

THE TWO SHAPES

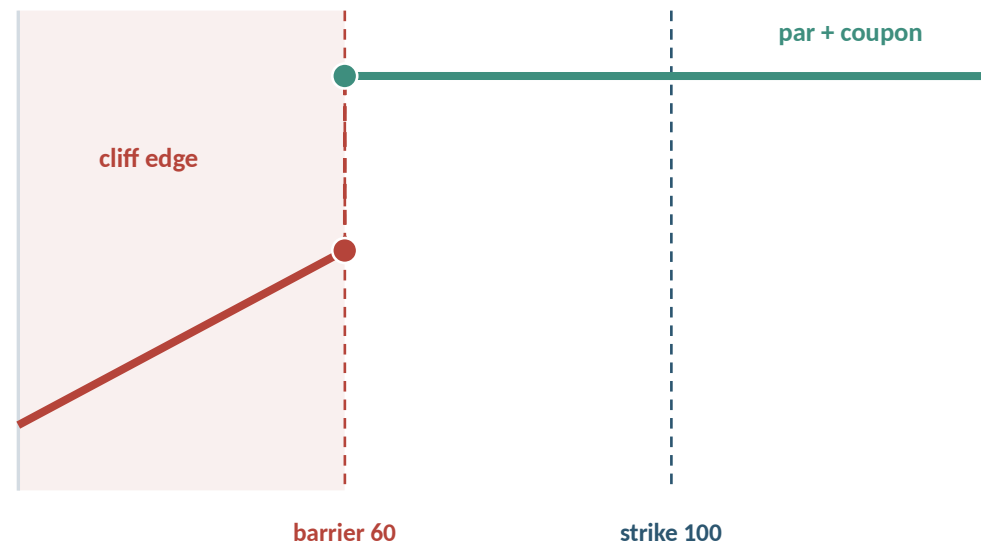
Adding a barrier changes the picture completely

Vanilla reverse convertible — strike at 100



Underlying at maturity (100 = initial)

Barrier reverse convertible — 60% knock-in



Underlying at maturity (100 = initial)

The barrier does not reduce the loss — it postpones it. One point below the barrier the client loses 40 points, instantly.

Is the coupon guaranteed, or does it have to be earned?

SSPA 1230

Unconditional

The coupon is paid on every payment date whatever the underlying does. The barrier affects redemption of principal only. This is the plain barrier reverse convertible, and it is still the Swiss convention.

SSPA 1255 / 1260

Conditional

Each coupon is paid only if the underlying is above a coupon barrier on that observation date. The client has sold an extra strip of digital options on top of the put.

- **The distinction is worth more than it looks.**
- A conditional coupon can stop precisely when the client most needs the income — in a drawdown — and the headline rate on the cover is the rate they will not be receiving.
- It is also a different product for us: an unconditional coupon is a fixed schedule, a conditional one is a schedule of contingent cashflows that has to be evaluated at every observation.
- Memory features exist to soften this. They are close to standard in Italy, common in Switzerland, and an optional extra in the US.

A conditional-coupon product is usually not booked under 1230. Check the category code as well as the name — they are set independently and both can be wrong.

TWO BARRIERS

The coupon barrier and the knock-in barrier price different risks

Coupon barrier

Observed on each coupon date. Decides whether that period's coupon is paid. Missing it costs income, not principal.

Knock-in barrier

Decides whether principal is repaid at par or the client takes the underlying's loss. Usually observed at maturity, sometimes continuously.

Same structure, two settings

Both at the same level

One number for the client to watch. The common case.

Payout Threshold below the Barrier

Coupons keep paying after principal protection has already gone.

Coupon barrier above the Barrier

Income stops long before principal is at risk. Buys a bigger headline.

- **Setting both at the same level is the dominant convention — one number for the client to watch.**
- They can sit either way round, and which way tells you what the structurer was optimising — a lower Payout Threshold protects the income stream, a higher one buys a bigger number on the cover.
- Check the observation convention separately for each barrier. Protection is often observed continuously while the coupon is tested once per period, at the same level.

A live Swiss note carries a continuously observed Barrier at 80% for principal and a Payout Threshold at 60% for the coupon — the second arrangement above, and the one that surprises people.

THE NAMING PROBLEM

One payoff, five names, and a code that means two things

Switzerland	Conditional Coupon Reverse Convertible, with or without a barrier	SSPA 1255 / 1260
United States	Contingent Income Auto-Callable, or Phoenix Autocallable	No formal category
Italy and France	Phoenix — as against Athena, which autocalls but pays no periodic coupon	EUSIPA 1260, Barrier Express
Germany	Folded into the Express-Zertifikat family	EUSIPA 1260
Korea	Step-down autocallable ELS	Statutory class, not a marketing name

The code 1260 means a conditional-coupon barrier reverse convertible in Switzerland and an express certificate everywhere else in Europe.

Same number, different product. On our own paperwork the two Swiss codes are used precisely: 1255 for a conditional coupon with no barrier, 1260 once a barrier is added.

WORKED EXAMPLE

A live US autocallable, term by term

Issuer	A bank; the client takes its credit risk for three years
Underlyings	Worst-of three indices — Russell 2000, Nasdaq-100, Nikkei 225
Tenor	3 years, with monthly observations
Coupon	0.958% per month ≈ 11.50% p.a., contingent
Coupon barrier	60% of each index's initial level
Autocall trigger	100% of initial, flat — no step-down
Non-call period	First call observation is roughly three months in
Downside threshold	60%, observed only at maturity (European)
At maturity	Above 60%: par. Below 60%: par × worst performer, so a 45% index means a 55% loss

The three questions

What did the client sell? A down-and-in put on the worst of three indices.

What makes it worse? Any one index falling 40%, and the three of them decoupling.

When does it end? Most likely at the first observation after a rally — not in three years.

11.50%

coupon p.a., contingent

−40%

before principal is lost

Terms from a Nomura Holdings 424B2 filing, issued 1 May 2026, US\$7.1m. Representative rather than exceptional.

WHAT EACH FEATURE COSTS

One share, one morning, nine structures

Twelve-month notes on the same share, priced the same day. Where there is a barrier it is at 80%. Only one thing changes at a time. Levels are indicative and rounded to a tenth of a point.

Reverse convertible, no barrier	11.6%	The base case. Principal is at risk from the first point below the strike.
+ continuous 80% barrier	10.7%	The barrier costs about nine tenths of a point. That is what conditional protection is worth.
barrier on the daily close instead	10.6%	Softer observation, so the barrier survives more often — and the client pays for it in coupon.
+ quarterly autocall at 100%	10.9%	Pays more than the plain barrier note: the client has sold the early-redemption optionality.
autocall stepping down 100 / 99 / 98 / 97	10.7%	Easier to redeem, so the expected life is shorter and the quoted rate eases back.
callable at the issuer's discretion	11.7%	Roughly a full point more than the mechanical autocall. Discretion is the most expensive feature here.
callable, quanto into a second currency	7.9%	Removing the currency risk costs nearly four points. Quanto is never free.
conditional coupon, no barrier • 1255	11.9%	The coupon is no longer guaranteed, so the headline rises.
conditional coupon with barrier • 1260	11.3%	Both features together. Compare it with the 11.6% base case and you can price each one.

These are all the same trade on the same share. The coupon differences are a price list for the features.

Callable: the issuer decides, and it is not doing you a favour

- **The issuer holds an option to redeem the note early, at par plus accrued coupon.**
- It exercises when the note has become expensive for it to keep — typically when volatility has fallen or the underlying has rallied.
- That is precisely when the client would rather keep receiving the coupon. Optionality that belongs to the issuer is optionality the client has sold.
- In exchange the client gets a higher headline coupon than the equivalent non-callable note. That gap is the price of the call.
- **Daily-callable variants exist in Hong Kong retail ELIs. The issuer can call on any business day, which makes the expected life very short.**

Callable

Issuer's discretion. Exercised in the issuer's interest, on its own model.

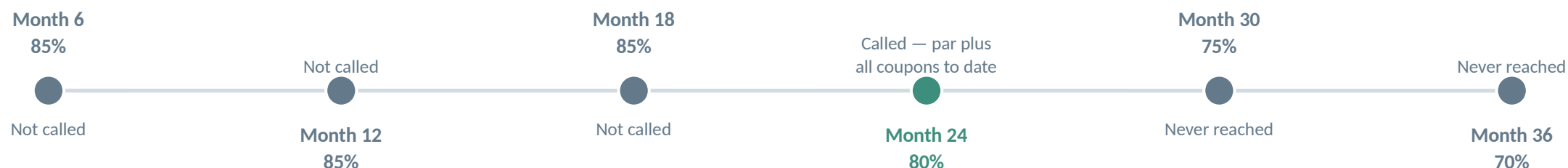
Autocallable

Mechanical. Triggered by an observable level, with no discretion on either side.

Clients almost always prefer autocallable, because 'the bank decides' is a hard sentence to put in a suitability file.

Autocallable: the structure that dominates the market

A step-down schedule makes the note progressively easier to redeem



Why it steps down

Each observation calls at or below the level of the last. The issuer wants the note gone; the client wants coupons. The schedule is a negotiated compromise.

Why the desk likes it

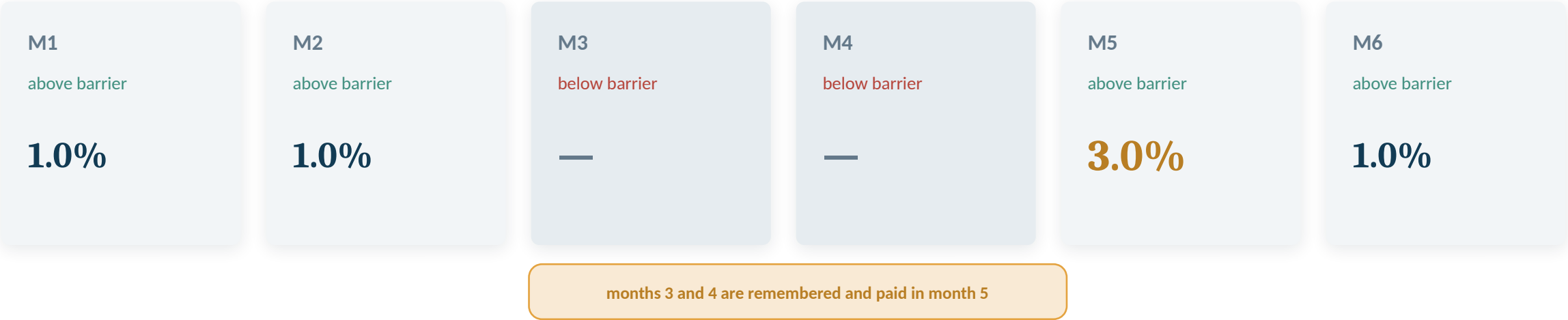
Rebalancing risk is spread across six trigger points instead of concentrated at one. Post-2015 'lizard' barriers do the same job.

Why it matters to us

A single trade generates six conditional cashflow branches. Get the observation calendar, business-day conventions and non-call window wrong and every downstream number is wrong.

Lock-in and memory: coupons that are deferred, not lost

Memory coupon — 1.0% per month, barrier at 80%



Memory coupon

In the documentation this is a Payout Postponement Event: the coupon is not lost, it is postponed and added to the next Payout Amount. Also called a snowball coupon.

Lock-in (bonus / capped)

Once the underlying trades above a level, a minimum return is banked for good and cannot be lost, whatever happens afterwards.

Why it sells

It removes the single most painful client conversation — 'you missed one observation by 0.3% and lost the coupon' — for a modest reduction in headline rate.

Worst-of: paying the client to sell dispersion

- Adding a name to the basket raises the coupon because it lowers the expected payout.
- The note references the weakest performer. A fourth name can only make the weakest performer weaker — never stronger.
- The worst-of option is worth most when the names decouple, because the worst performer is then further from the pack. The client has sold that dispersion; the desk, which now owns the option, is the side that is short correlation.
- That matters because correlation rises in a crash — the desk's position deteriorates exactly when the market is moving fastest. Desks recycle the exposure through correlation and covariance swaps.
- Two to four names is the norm. Three is the most common. Every additional name is more coupon and materially more risk.

A three-name worst-of at maturity

Nvidia +42%

Microsoft +9%

Tesla -48%

Barrier at 60%

Tesla at 52% of initial is through the barrier.

The note redeems at 52% of par.

The other two names are irrelevant.

Dual currency investments: the same trade in FX

Structure	Deposit in a base currency + short FX put. The premium is paid out as enhanced interest
At maturity	If the alternate currency has not weakened through the strike: repaid in base currency with the enhanced rate
Otherwise	Principal converted at the strike rate into the weaker currency — so the client is long the currency that just fell
Tenor	1 week to 6 months. Far shorter than equity structures
Pairs	Any combination of HKD, USD, CNY, AUD, NZD, GBP, CAD, EUR, JPY
Minimum	Around HK\$50,000 equivalent at a Hong Kong retail bank
Principal protection	None, in base-currency terms. Banks state this explicitly on the product page

Two fields, side by side

A one-month FX note shows Coupon 11.7% p.a. and, three lines below it, Maximum Yield 1.0%. Both are correct. Always read the second number.

Where it goes wrong

Clients roll these repeatedly, collecting small premiums, until one conversion leaves them holding a currency they did not want at a rate they cannot recover from. The Taiwanese target-redemption forward crisis was this pattern with leverage bolted on.

Typical terms, and how much they vary by market

Tenor	1–3 years in the US and Europe. 3 years standard in Korea. Under 1 year for 98% of Taiwanese trades
Coupon	Roughly 8–12% p.a. on an index worst-of; 20%+ on a volatile single-stock worst-of
Knock-in barrier	60–65% US and Europe. 50% typical for Korean index ELS. 35–45% on high-volatility single stocks
Observation of barrier	Increasingly European — observed only at maturity. A real de-risking versus the continuous barriers of the 2015 vintage
Autocall trigger	100% flat is common in the US and Switzerland; gentle Swiss step-downs run 100/99/98/97, Korean ELS 85/85/85/80/75/70
Observation frequency	Monthly or quarterly for index notes; semi-annual for Korean single-stock ELS
Non-call period	Three to six months from trade date
Basket size	2–4 names, three being the modal US structure

Peak vega: the hedge that turns against you on the way down

Dealer vega on an autocallable book



The flip

As spot falls, autocall probability collapses, expected life extends, and the book gets longer vega — until the barrier comes into view and the embedded put becomes near-certain. Then vega falls away sharply.

The crowd

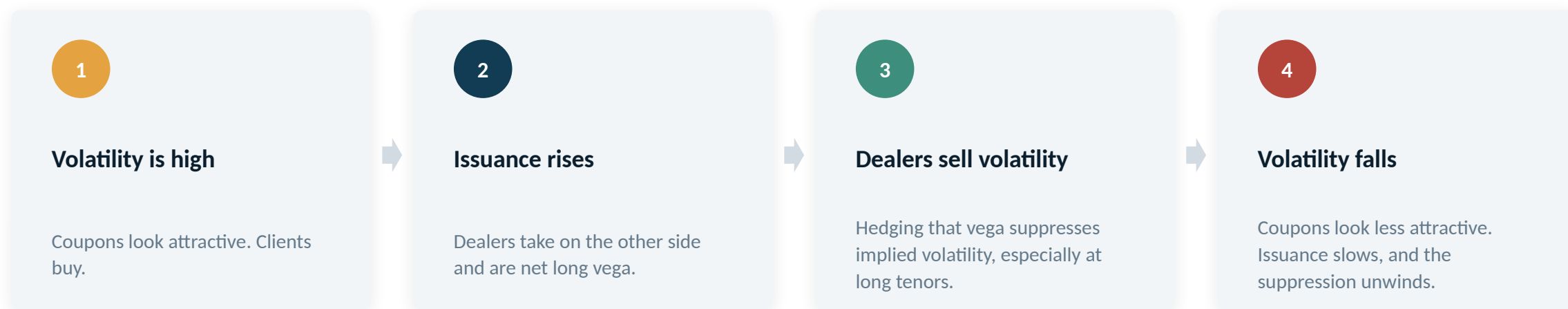
Every dealer issues on the same handful of indices at similar levels in the same vintage. The flip happens to all of them at once, in the same direction, in a market with no liquidity.

The precedents

Korea 2015 ($\approx$ US\$300m of street losses), Natixis Q4 2018 (€260m), and March 2020, when European dividend cancellations added an unhedged loss on top of peak vega.

Curve shown is a schematic of the shape the desk describes, not a calibrated model output.

Autocall books supply the market's long-dated volatility



When Japan and Korea curbed retail structured product sales, long-dated Nikkei and HSCEI options lost their main source of supply and implied volatility rose.

This is why the desk cares about regulation in markets it does not sell into. Issuance in one jurisdiction sets the price of volatility for everyone. Hong Kong is now a larger supplier of structured-product vega than Korea or Japan.

Korea 2024: the largest retail loss on record

KRW 19.3tn

HSCEI-linked ELS outstanding

As at November 2023; 82% sold through five commercial banks

−58%

HSCEI drawdown from strike

From above 12,000 in 2021 to about 5,100 in early 2024

KRW 4.6tn

realised investor losses

Roughly US\$3.2bn, concentrated in the first half of 2024

20–65%

compensation ratios

Set by the FSS in March and May 2024, varying by bank

- **The product did what it said it would. The distribution did not.**
- Investors bought in 2021 with the index near its peak. Three-year notes matured in 2024 after the index had roughly halved — well through 50–65% knock-in barriers.
- The regulator's finding was about sales, not structure: 90.6% of bank sales were in-branch, and ELS counters were not separated from deposit counters, so clients conflated the two.
- Remedies: bank ELS sales restricted to 5–10% of branches with physically separated areas; sales resumed September 2025; a fine of roughly KRW 600bn across five banks was reported in June 2026, still subject to approval.
- From August 2026, a mandatory alert fires when an underlying comes within 10 percentage points of the knock-in barrier.

Five things about reverse convertibles

- 1 It is a bond plus a short put. The coupon is the put premium, spread out over time.
- 2 A barrier postpones the loss, it does not shrink it. One point through the barrier costs 40 points.
- 3 Callable means the issuer chooses; autocallable means a level chooses. Clients should know which they own.
- 4 Every extra name in a worst-of raises the coupon and lowers the expected payout. That is not a coincidence.
- 5 The desk's risk is not the mirror of the client's. Peak vega, correlation and dividends fail on their own schedule.

Next: Deck 3 — the Asian payoffs. Fixed coupon notes and equity linked notes.

FCN & ELN

Fixed coupon notes and equity linked notes — the short-dated, worst-of, physically-settled products that dominate Asian private banking flow.

START HERE

The same economics, five different names

FCN

Everywhere in Asian PB

Fixed Coupon Note. Unconditional coupon, knock-in barrier, autocall, worst-of basket, physical settlement into the worst performer.

ELN

HK, SG and Swiss PB

Equity Linked Note. Buy at a discount, get shares if the stock closes below strike. One observation at maturity — and on a Swiss termsheet it is classified as a Discount Certificate, SSPA 1200.

ELI

HK retail (SFC-authorised)

Equity Linked Investment. The regulated retail wrapper. Economically an ELN, but a different regulatory perimeter — and modern ELIs now carry autocall and 'airbag' features too.

ELS

Korea

Equity-Linked Securities. A statutory class, not a marketing name. Long-dated (3y), mass retail, mostly index-linked, cash-settled, step-down autocall.

Uridashi

Japan

The wrapper term for the structured bonds sold to Japanese retail. FCN/ELN vocabulary is simply not used there.

Nobody polices these names. Read the term sheet, not the label.

A fixed coupon note, term by term

Underlyings	Worst-of two or three names — often HK tech, US megacaps, or Taiwanese semis
Tenor	3, 6 or 9 months. Six is the norm
Coupon	Unconditional. Paid on every coupon date regardless of where the shares are
Strike	Typically 90–100% of initial. This is the price at which the client may be delivered shares
Knock-in barrier	Commonly 60–80% of initial, observed daily or at maturity. Deeper on volatile single names
Knock-out / autocall	Often at or slightly below 100% of initial, observed daily or monthly after a non-call window
If autocalled	Note redeems at par, coupons accrued to date are paid, and the trade is over
If knocked in and below strike	Physical delivery of the worst performer at the strike price
If neither	Par plus all coupons. The best outcome, and the least likely on a volatile basket

The coupon is the trap

The word 'fixed' is true and irrelevant. It describes the coupon, never the principal.

A client can collect every scheduled coupon in full and still lose 40% of their money at maturity, because the two are unrelated.

This is the single most common misunderstanding in the product set, and it is the one that shows up in every mis-selling file.

Compare with Deck 2: an FCN is a reverse convertible with an unconditional coupon and physical settlement bolted on.

WORKED EXAMPLE

Six months, three names, four possible endings

Worst-of Alibaba / Tencent / Xiaomi • 6 months • 12% p.a. paid monthly • strike 90% • knock-in 70% • autocall 100% • notional US\$1m

Autocalled in month 2

US\$1,020,000

All three names above 100% on the month-2 observation.

Par plus two monthly coupons. Annualised return 12%, realised return 2%.

Runs to maturity, no knock-in

US\$1,060,000

Nothing ever traded below 70%. Worst name ends at 84%.

Par plus six coupons. The client keeps everything.

Knocked in, recovers

US\$1,060,000

Xiaomi touched 68% in month 3 but the worst name ends at 94% — above the 90% strike.

Knock-in is necessary but not sufficient. The final level still saved it.

Knocked in, ends low

US\$671,000

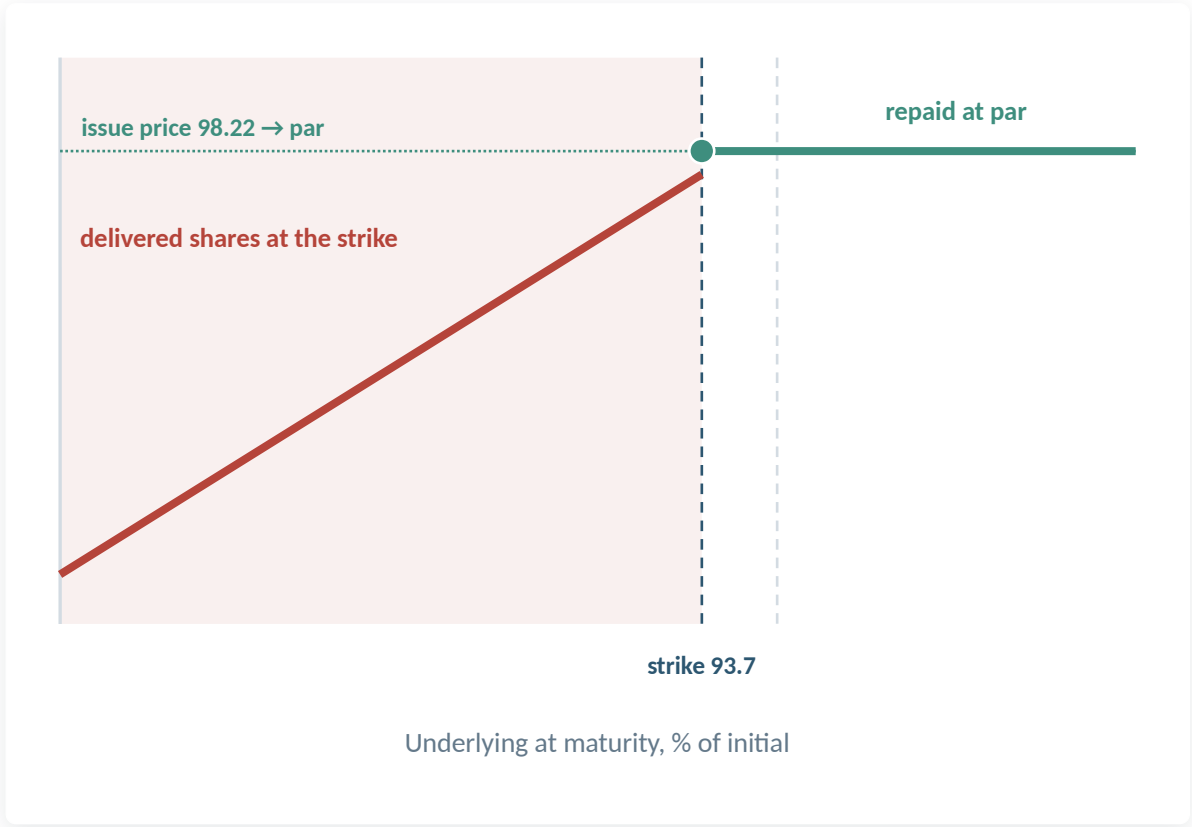
Xiaomi ends at 55%. Below the barrier and below the strike.

Delivered the worst performer at the 90% strike — worth US\$611,000 — plus US\$60,000 of coupons. Loss 33%.

Illustrative terms and arithmetic. Coupon 1% per month; the delivered share count is notional $\div$ (90% $\times$ initial price). Check live levels with the desk.

An equity linked note is a discount certificate

Client return per 100 invested



Underlying	Henderson Land (0012.HK)
Spot at issue	HK\$64.50
Strike	HK\$60.45 — 93.7% of spot
Issue price	98.22% of par
Tenor	51 days
Annualised yield	13.70%
Breakeven	HK\$59.37

At maturity	≥ strike: repaid at par. < strike: delivered shares at HK\$60.45
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The classification is not a quirk. A Swiss ELN termsheet carries the code 1200 — Discount Certificate — because that is exactly what the payoff is. Deck 4 takes the same instrument apart from the other direction.

SIDE BY SIDE

Three products, one family

ELN / ELI

Tenor

1–6 months

Coupon

None — bought at a discount

Observations

Usually one, at maturity

Basket

Usually single name

Barrier

Often none

Settlement

Physical

Complexity

Low

FCN

Tenor

3–9 months

Coupon

Unconditional, monthly

Observations

Daily or monthly

Basket

Worst-of 1–4

Barrier

Knock-in plus autocall

Settlement

Physical

Complexity

High

Reverse convertible

Tenor

1–3 years

Coupon

Often conditional

Observations

Monthly or quarterly

Basket

Worst-of 2–4

Barrier

Knock-in plus autocall

Settlement

Usually cash

Complexity

High

Same risk, different packaging, different regulatory perimeter. The ELN column describes the classic form — modern HK retail ELIs often add autocall and airbag features.

Physical settlement is not a detail

- **A US autocallable pays cash. An Asian FCN hands over shares.**
- The client ends up holding a specific number of a specific stock, in a specific market, on a specific settlement date — bought at a strike set six months earlier.
- Share counts have to be computed and rounded, and the residual paid in cash. Rounding conventions differ by issuer and are a genuine source of breaks.
- The delivered stock has to settle in its home market, under its own settlement cycle, holiday calendar and stamp duty regime. Hong Kong and US names are not usually mixed in one basket for exactly this reason.
- Corporate actions between strike date and delivery — splits, rights issues, spin-offs — change the share count and the strike. Somebody has to adjust them.
- **For the development team this is where most of the real work lives. The payoff is arithmetic; the delivery is operations.**

Share count

$\text{notional} \div (\text{strike \%} \times \text{initial price})$, rounded down to the board lot

Residual cash

the rounding remainder, paid alongside the shares

Settlement market

the underlying's home exchange, not the note's currency

Corporate actions

adjust strike and quantity, or the whole trade is wrong

The same product behaves differently in each jurisdiction

Hong Kong

Now the largest source of structured-product vega in the region — roughly triple its 2024 level, and ahead of Korea and Japan. Alibaba and Tencent are the anchor names. Retail ELIs sit alongside PI-only ELNs.

Singapore

Private-bank driven, US tech and AI themes, gold gaining ground. Structured note volumes reported up around 50% year on year in the first half of 2026.

Taiwan

98% of trades one year or shorter, almost entirely plain FCNs. Little product innovation, and a regulator that has been widening rather than narrowing the eligible investor base.

Korea

A different animal: three-year, index-linked, cash-settled ELS sold to mass retail. Post-2024 the mix has shifted to roughly half principal-protected, and distribution has moved from banks to brokerages.

TYPICAL TERMS

What the desk is actually quoting

Tenor	3–9 months for FCNs; 1–6 months for ELNs; 3 years for Korean ELS
Coupon	Roughly 7–15% p.a. on a private-bank FCN; higher on volatile single-stock worst-of baskets
Strike	Retail examples cluster at 90–100% of initial; institutional worst-of baskets are struck lower
Knock-in	Commonly 60–80% of initial; published retail examples sit higher, institutional worst-of baskets lower
Knock-out	At or slightly below 100%. Autocall below 100% is common in Asia and unusual in the US
Basket size	1–4 names. HK and US stocks are generally not mixed in the same basket
Settlement	Physical delivery of the worst performer, at the strike
Underlyings	Alibaba, Tencent, Meituan, Xiaomi; Nvidia, Microsoft, Tesla; Samsung Electronics, SK hynix, TSMC

Published retail examples sit at gentler levels than institutional worst-of trades. Check the desk's own tape before quoting a range.

Four regulators, four different answers

Hong Kong

SFC / HKMA

- PI threshold: HK\$8m portfolio for individuals
- ELIs and ELNs are classified as complex products
- SPI regime since July 2023 streamlines transaction-level suitability at HK\$40m portfolio or HK\$80m net assets
- Annual rather than per-trade complex-product warnings for SPIs

Singapore

MAS

- Accredited Investor: S\$300k income, S\$2m net personal assets, or S\$1m net financial assets — opt-in since 2019
- 2026 reform renames SIP/EIP to 'complex' and 'non-complex'
- CKA and CAR merged into one assessment that no longer expires
- Mandatory advice removed for most investors; retained for 'Selected Clients'

Taiwan

FSC

- High-asset customer regime at NT\$100m
- April 2026 amendments widened the eligible buyer base for offshore structured products to professional juristic persons and professional natural persons
- Direction of travel is liberalising, not tightening

Two episodes that shaped how these are sold

March–April 2020

FCN losses across Asian private banks

Coupons proved no defence against a worst-of drawdown. What turned a bad quarter into a disaster was leverage: notes were financed at over 50% loan-to-value on average, so the fall triggered margin calls and forced unwinds at steep discounts, well before maturity.

“In a bull market, investors keep collecting coupons on these notes and they feel it's a great investment. When the market turns, they get stuck with unimaginable losses.”

2024

Korea's HSCEI-linked ELS

KRW 4.6 trillion — about US\$3.2bn — of realised retail losses on notes sold in 2021 and maturing after the index roughly halved. The regulator's finding was about distribution: over 90% of bank sales were in-branch, with ELS counters not separated from deposit counters.

- Compensation set at 20–65% of losses, by bank
- Bank sales confined to 5–10% of branches
- Roughly 97% of cases settled voluntarily by early 2026
- Knock-in proximity alerts mandatory from August 2026

Where these products actually break

1 Calendars

Every observation is a date in a specific market with its own holidays. A fixing on a non-business day has to roll — and which way it rolls is a contractual term.

2 Disrupted days

Trading halts, suspensions and limit moves defer a fixing. The fallback logic is in the documentation and it is rarely simple.

3 Barrier semantics

Daily close versus intraday touch versus final-fixing-only. Three different products, one word on the term sheet.

4 Worst-of resolution

Performance is measured per name against its own initial level. Ties, corporate actions and different currencies all need explicit handling.

5 Early termination

Autocall means the position ends mid-schedule. Accruals, coupon proration and downstream reporting all have to cope.

6 Physical delivery

Share counts, board lots, residual cash, settlement markets and stamp duty. The messiest part of the lifecycle by some distance.

Five things about the Asian payoffs

- 1 'Fixed coupon' describes the coupon and nothing else. The principal is fully at risk.
- 2 An ELN is one short put over a few weeks. An FCN is that plus a barrier, an autocall and a basket.
- 3 The names are not standardised. HK says ELI, Korea says ELS, Japan says uridashi — read the terms.
- 4 Physical settlement is the defining Asian feature, and it is where the operational risk lives.
- 5 Both major loss episodes came from distribution and leverage, not from the payoff misbehaving.

End of the series. The interactive payoff explorer covers all four decks — open it alongside your first term sheet.

Discount & bonus

The European listed side of the book — where the option premium arrives as a price discount or a conditional bonus instead of a coupon.

Two families, and the line between them

12xx

Yield enhancement

Capped upside in exchange for a known return. Discount certificates, barrier discounts, reverse convertibles, BRCs.

13xx

Participation

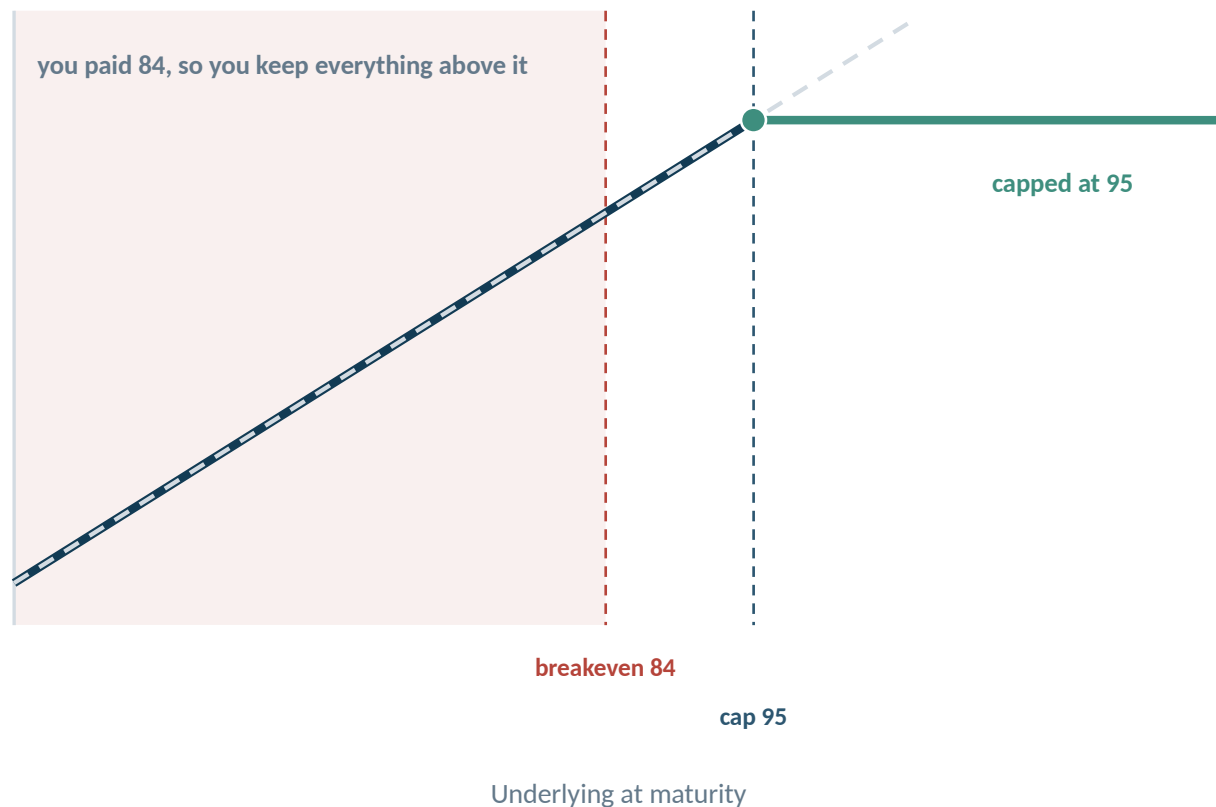
Open-ended upside. Trackers, outperformance, bonus certificates, twin-win.

The test is not how the product is built — it is whether the upside is open-ended. A cap moves a product across the family line even though only one option leg changed.

A capped bonus certificate whose cap equals its bonus level is economically the same instrument as a barrier discount certificate. Both pay a fixed maximum if the barrier survives and both convert into the underlying if it does not. Same payoff, two names, two families.

Buy the underlying cheaper, give up everything above the cap

Redemption per certificate, underlying at 100, cap 95, purchase price 84



Above the cap

Redeem at 95. You bought at 84, so the return is 13.1% — and not a penny more, however far the underlying runs.

Between 84 and 95

Redeem at the underlying's value. Still a profit, because you paid 84.

Below 84

A loss — but always 16 points better than owning the share outright.

The dashed line is a direct holding in the share. The certificate beats it everywhere except a rally through the cap — which is exactly the trade being made.

One instrument, two equally correct descriptions

The covered-call form

Zero-strike call on the underlying + short call struck at the cap

This is how the German and Swiss listed market describes it. The zero-strike call is used rather than the share itself because the investor does not receive dividends — and those forgone dividends are a large part of what funds the discount.

The bond-plus-put form

Zero-coupon bond redeeming at the cap + short put struck at the cap

This is the SSPA formulation, and the one that matters for us: it makes the identity with a reverse convertible immediate.

Put-call parity, nothing more. Do not let anyone leave thinking these are two different risks.

A discount certificate is a reverse convertible paid up front

	Discount certificate	Reverse convertible
How the premium reaches the client	As a price discount, on day one	As a coupon, spread over the life
Purchase price	Below spot	Around par
Coupon	None	Yes
Upside limit	The cap, which is also the option strike	Par plus coupon
Downside trigger	No barrier — exposure starts immediately below the cap	Below the strike, or below the barrier for a BRC
Cushion	Small and certain — the discount	Large and conditional — the barrier
Max redemption	The cap amount	Par plus coupon
EUSIPA family	Yield enhancement, 1200 / 1210	Yield enhancement, 1220 / 1230

Same strike, same maturity, same risk. Neither is the safer one.

The typical discount certificate is not a slightly capped share

22.4%

average discount

At first observation, across the German listed market

-14.3%

average distance to cap

The underlying already trades above the cap

78%

issued in the money

Started with the underlying above the cap level

332 d

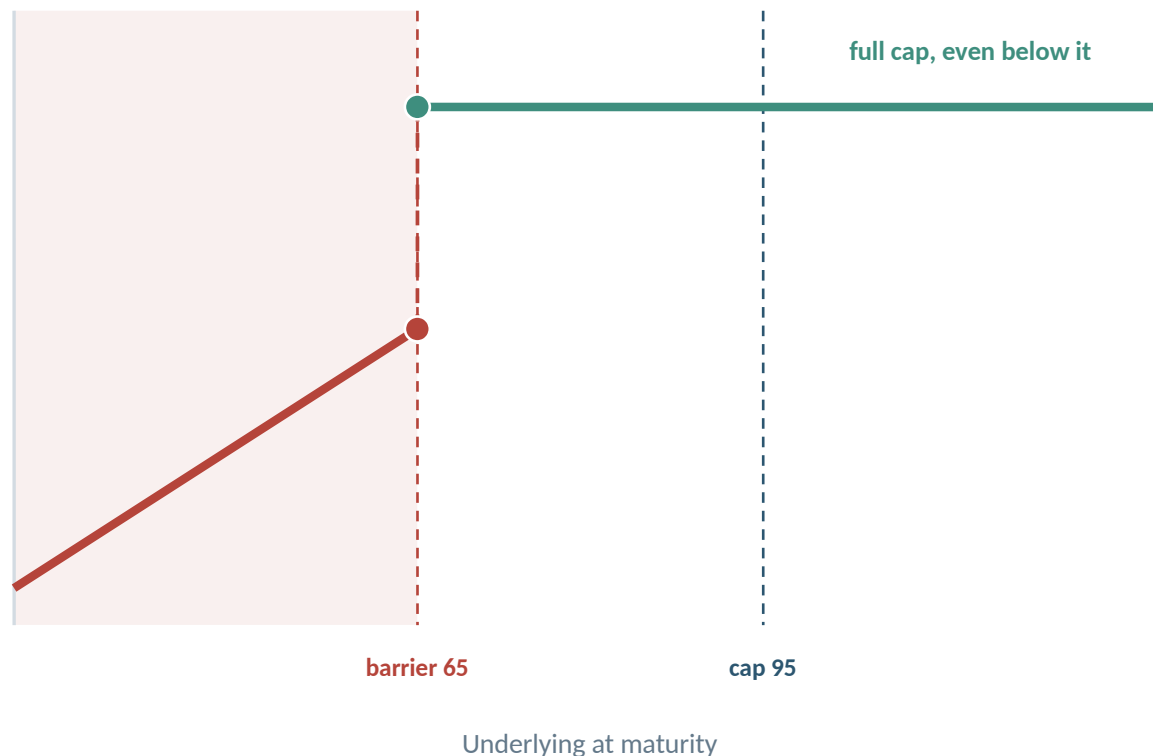
average remaining term

Short-dated, high turnover, frequently rolled

- **Read those four numbers together and the product changes character completely.**
- A certificate issued with the underlying already above its cap has no upside participation at all. It is bought for the buffer, not for the equity exposure.
- So the realistic outcome distribution is: collect the capped return most of the time, with an equity tail. It behaves far more like credit than like a share.
- 73% reference single stocks, 27% indices. Settlement is physical for German and most European single names, cash for indices, commodities and US, Spanish, Finnish and Swiss shares.
- Tenor runs 3 months to 5 years on paper; 6 to 24 months is where the volume is.

Barrier discount: a conditional cushion instead of a certain one

Cap 95, barrier 65, purchase price 88



- If the barrier is never touched, the client gets the full cap amount — even if the underlying finishes below the cap.
- That is a genuinely better payoff than a plain discount certificate in the middle of the range, and it is paid for with a smaller discount up front.
- If the barrier is touched, the barrier feature vanishes and it reverts to a plain discount certificate.
- Economically: zero-coupon bond plus a short down-and-in put. The same structure as a barrier reverse convertible, with the premium taken as a discount rather than a coupon.
- **Worst-of versions are standard and buy a bigger discount or a deeper barrier, at the usual price — the client is short dispersion.**

A dual currency deposit, wrapped as a security

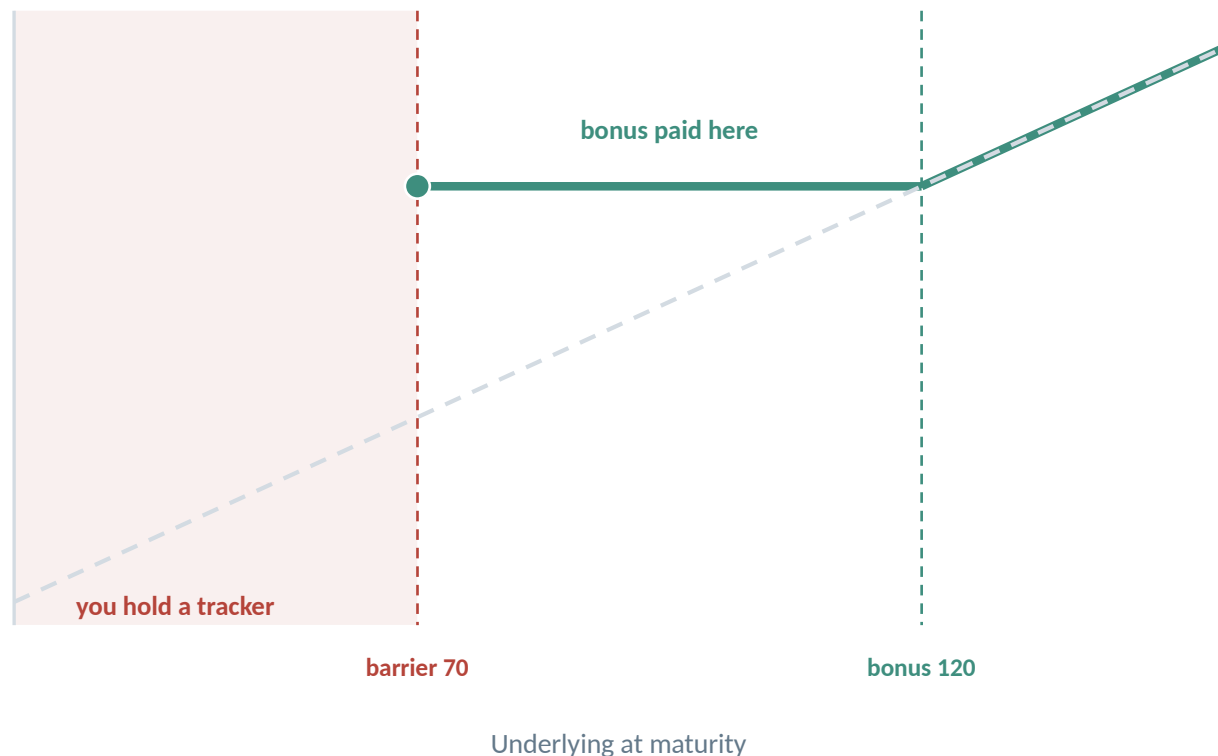
	FX discount certificate	Dual currency investment
Legal form	A security with an ISIN, listable and transferable	A deposit or OTC contract with the bank, not transferable
Return delivery	An up-front discount to the reference rate	An enhanced interest rate on the deposit
Conversion	Redemption in the alternate currency if the pair finishes the wrong side of the cap	Principal converted at the strike if the pair finishes the wrong side
Classification	SSPA 1200, yield enhancement	Not in the SSPA or EUSIPA taxonomy at all
Credit exposure	Unsecured note — issuer credit risk	Bank deposit, which may rank differently in insolvency
Typical tenor	Weeks to about two years	Days to three months

The same short-FX-option trade the treasury desk sells as a dual currency deposit, wrapped so it can be booked and held in a securities account.

Live Swiss examples are small: one listed USD/CHF discount certificate maturing February 2026 had a total issue size of USD 56,000. Treat FX discounts as a bespoke corner, not a volume business.

A floor if the barrier holds, and the upside stays open

Bonus level 120, barrier 70, underlying at 100 on the trade date



Above 120, barrier intact

Full 1:1 participation. The bonus level is irrelevant here — you simply get the underlying. Upside is uncapped.

Between 70 and 120, barrier intact

You receive 120. This is the whole point of the product: a positive return from a flat or moderately falling market.

Barrier touched at any point

The bonus is extinguished permanently and does not revive on a recovery. You now hold a tracker — and a tracker that still pays you no dividends.

The client pays for it with their own dividends

Long zero-strike call

Tracks the underlying from a strike of zero. It gives the uncapped upside, and the tracker profile after a breach.

Long down-and-out put struck at the bonus level

Its knock-out barrier is the certificate's barrier. If the barrier survives, this put lifts redemption up to the bonus level. If it knocks out, only the zero-strike call is left.

A zero-strike call costs spot minus the present value of dividends, because its holder never receives them. That gap is the entire budget that buys the put.

What follows from that

High-dividend names price better

A bigger dividend stream is a bigger budget, so it buys a higher bonus level, a deeper barrier, or both.

Zero-dividend names barely work

With almost no budget you cannot build a meaningful bonus without adding a cap or a worst-of basket.

Withholding tax bites

The issuer usually cannot price in the full gross dividend, so the budget is smaller than the headline yield suggests.

The bonus is not a gift

It is the client's own dividend stream, repackaged as conditional downside protection.

After a breach you are worse off than owning the share

Touch the barrier once and the bonus is gone for good. What you are left holding is a tracker on the underlying that still pays you no dividends — because the issuer already spent them.

It does not matter that the underlying recovers the next day, or that it finishes above the bonus level. The down-and-out put knocked out, and knocked-out means gone.

Permanent, not temporary

There is no recovery mechanism, no memory feature, no second chance. This is the single most common client complaint on the product.

Continuous observation is punishing

On a listed European bonus certificate one intraday print through the barrier is enough. An overnight gap or a flash move does it.

You still do not get the dividends

The zero-strike call never entitled you to them. So post-breach you underperform a direct holding by the full dividend yield, every year, to maturity.

Ask a new joiner what a breached bonus certificate is worth relative to the share. If they say 'the same', they have missed the dividends.

Five ways to change the bonus certificate

Capped bonus

Add a short call at the cap, usually set at the bonus level. Redemption is then exactly the bonus whenever the barrier survives. The premium buys a higher bonus, a deeper barrier, or both.

Reverse bonus

Inverted: the barrier sits above spot and the payoff rises as the underlying falls. Total loss if it finishes at or above the reverse level. A bearish instrument, and priced off borrow rather than dividends.

Bonus pro

The barrier is only monitored over a window ending at maturity, not from day one. Far lower knock-out probability, so weaker headline terms. Beware — issuers also use the name for multi-barrier designs.

Worst-of bonus

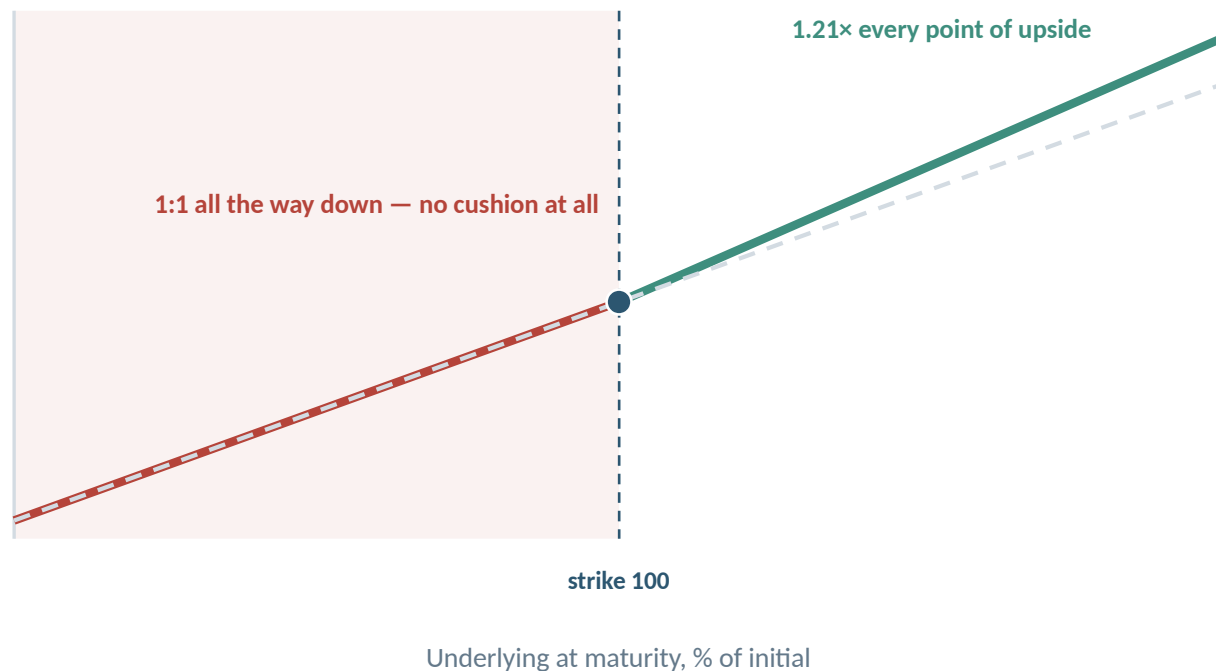
Barrier and bonus reference a basket; any one name breaching kills it, and after a breach you track the worst performer. Much cheaper protection, at the usual price in correlation.

Outperformance bonus • 1330

A bonus floor plus leveraged upside above the strike. You are buying two features from one dividend budget, so both are weaker than the single-feature versions.

Outperformance: the same budget, aimed upward

Strike 100%, participation 121%, physical settlement below the strike



Above the strike

The client receives the strike plus 1.21 times every point of gain above it. Uncapped.

Below the strike

Physical delivery of the underlying. One-for-one with the fall, exactly like owning the share — but without the dividends.

Where the leverage comes from

The same forgone dividend stream that funds a bonus certificate. Here it buys an extra call struck at the strike instead of a down-and-out put.

A bonus certificate buys a floor with the dividends. An outperformance certificate buys leverage.

Bonus Enhanced Notes: read the term sheet, not the name

Underlyings	Worst-of two or three single names
Tenor	1 to 12 months; six is the norm
Coupon	A flat, non-annualised bonus — 9.18% over six months in the desk's reference structure
Coupon barrier	100% of initial. Every name must be at or above its initial level
Put strike	90% of initial
Knock-out / autocall	None. The note runs to maturity
Upside participation	None. The bonus is the maximum
Settlement	Physical delivery of the worst performer at the 90% strike
Distribution	Private banks in Singapore and Malaysia, at high account minimums

Why this slide exists

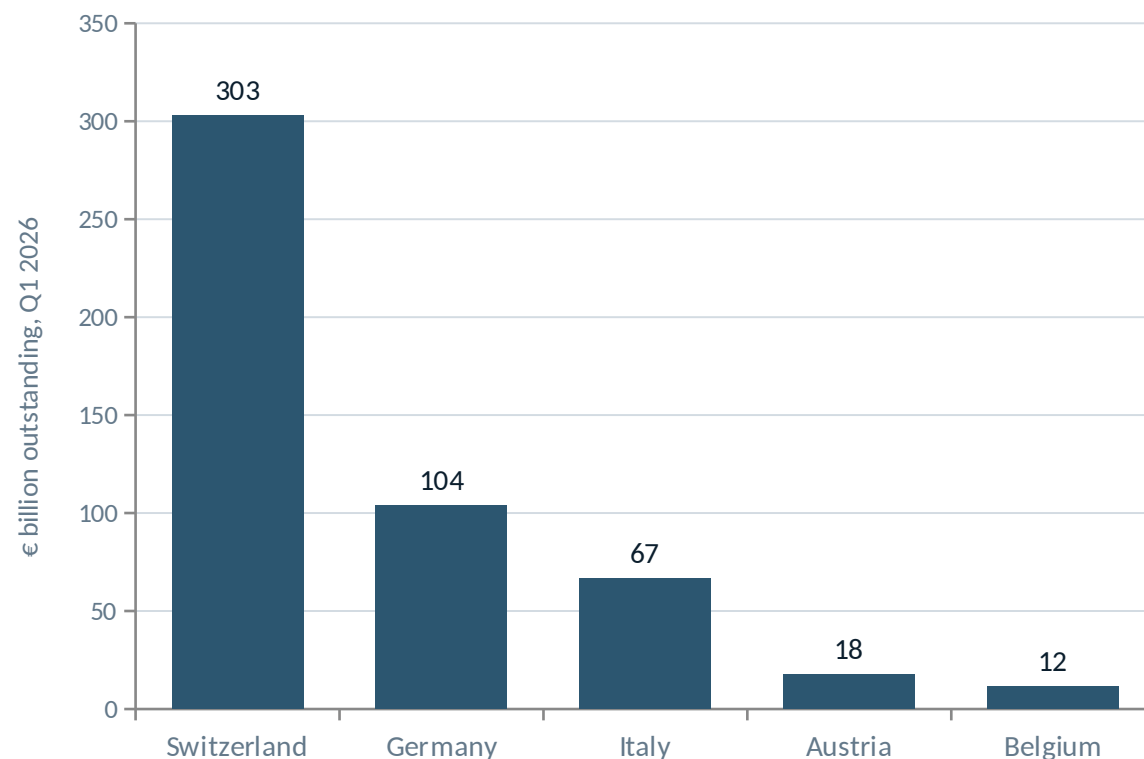
The name says bonus. The payoff is yield enhancement — a capped coupon, no participation, and physical delivery below the strike. That is a barrier reverse convertible without the autocall, not a bonus certificate.

There is no issuer term sheet for a product called a 'Bonus Enhanced Note'. Every published description is a distributor's explainer, and different distributors describe materially different payoffs under the same name — one uncapped and participation-like, one capped like ours.

It appears in no EUSIPA or SSPA taxonomy. If you must map it, ours maps to 1230 — barrier reverse convertible.

Relative scale, and why Zurich and Frankfurt are different businesses

European structured investment products by market



- **Switzerland is roughly 60% of European outstandings — but it is mostly OTC private-bank notes, not listed retail paper.**
- Germany is the opposite: an enormous number of listed instruments, dominated by fixed-rate notes and express certificates.
- In the German market, discount certificates are €4.2bn and bonus certificates only €1.2bn — about 4% and 1% of investment products.
- In Switzerland, yield enhancement is half of all turnover, with reverse convertibles at CHF 19bn a quarter against CHF 6bn for discount products.
- **So 'the European structured products market' means something completely different depending on which city you are standing in.**

Five things about discount and bonus

- 1 A discount certificate is a reverse convertible paid up front. Same strike, same maturity, same risk.
- 2 The cap is the option strike, not a barrier. The cushion is small and certain, not large and conditional.
- 3 The typical discount certificate is issued in the money. It behaves like credit with an equity tail.
- 4 A bonus certificate is funded by the client's own forgone dividends — and after a breach they lose those too.
- 5 A cap moves a product across a family boundary. And 'Bonus Enhanced Note' is a sales label, not a product type.

Next: Deck 5 — capital protected notes and mini futures, the two ends of the risk spectrum.

Protection & leverage

Capital protected notes at one end, mini futures at the other.

Opposite ends of the risk spectrum — and the same unsecured claim underneath.

WHERE WE ARE

The far ends of the map

Capital protection • 11xx

Capital protected notes

Most of the money buys a bond. What is left buys an option. The client cannot lose principal at maturity — provided the issuer is still there.

Leverage • 22xx

Mini futures

Almost none of the money buys anything except exposure. A few percent of adverse movement ends the position.

Both are unsecured senior claims on the issuer. The protected-note buyer thinks they bought safety and bought issuer credit; the mini-future buyer thinks they bought a futures position and bought issuer credit.

Same balance sheet, opposite ends of the risk spectrum, identical failure mode in a default. Lehman proved both halves of that sentence in 2008.

There is no collateral, no segregated pool and no deposit protection behind either product. That is the one thing they have in common, and it is the thing clients least expect.

Buy the floor first, then spend what is left

1

Buy the bond floor

To repay 100 at maturity, set aside the present value of 100 discounted at the issuer's own funding curve. At 4.3% over five years that is about 81.

2

Spend the rest on optionality

Option budget = 100 – floor – fees. At those levels, roughly 19 points, less costs.

3

Divide to get participation

Participation rate = option budget ÷ the cost of one unit of the call. Nineteen points buys about 0.96 of an at-the-money five-year index call.

Note the discount curve

The floor is discounted at the issuer's funding curve, not at the risk-free rate. A weaker issuer discounts harder, so its floor costs less and its option budget is bigger.

That is why a lower-rated issuer can advertise a better participation rate on identical terms. The client is not being given more upside — they are being paid for taking more credit risk, and usually not being told so.

If two CPNs on the same index look different, check the issuer before you check the structure.

Participation is a residual, not a feature.

THE FOUR LEVERS

Four ways to spend the same budget

Lower the protection

100% → 90%

A smaller floor to buy frees roughly eight points of budget at current rates. The client takes the first 10% of loss.

Lengthen the tenor

5y → 10y

A deeper discount means a cheaper floor. But the client is locked up twice as long and takes twice the duration risk.

Add a cap

uncapped → capped

The budget now buys a call spread rather than a call, which is materially cheaper. A 41.5% cap over five years roughly halves the option cost.

Change the underlying

index → futures index

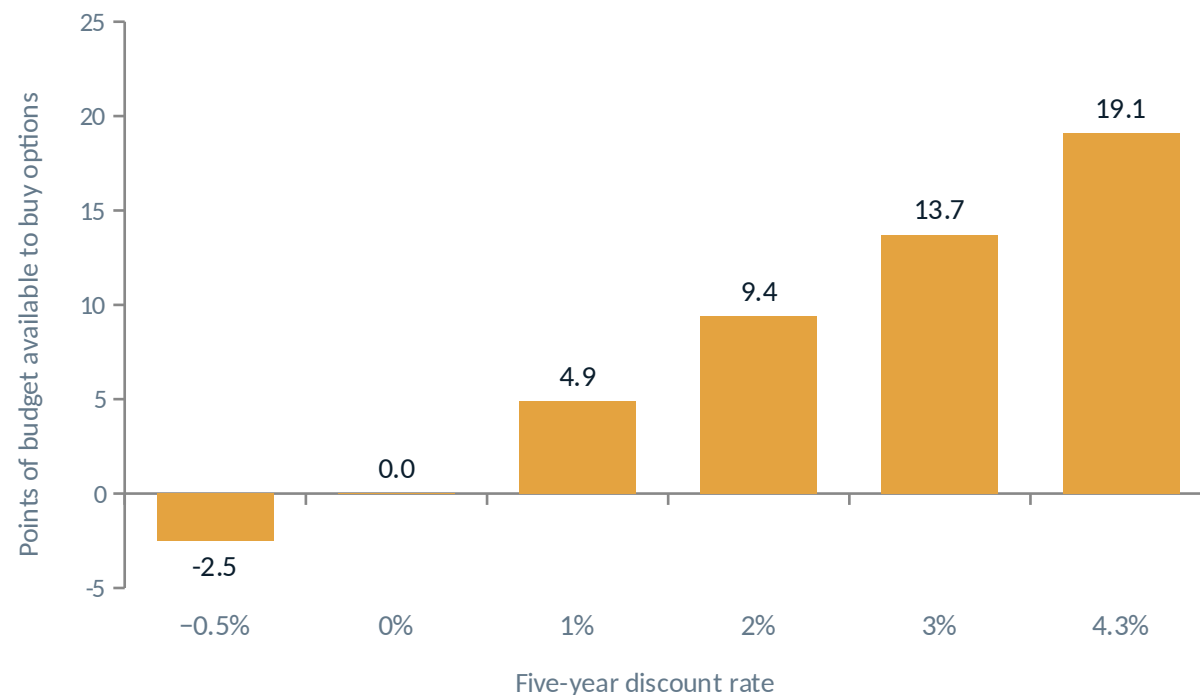
An excess-return or futures index strips out dividends and funding, so the same call is cheaper. This is why protected notes so often reference a futures index rather than the price index.

Four ways of spending one budget — pull any of them and the others have to move.

WHY THIS PRODUCT DISAPPEARED AND CAME BACK

At zero rates a protected note is not expensive — it is unbuildable

What is left over after buying a floor that repays 100 in five years



- **Add one to two points of distribution fees and the budget is negative well before rates reach zero.**
- That is the whole story of the 2015–2021 period. In France, before the ECB began tightening in 2022, roughly 95% of structured product issuance was in non-guaranteed equity structures — autocalls and barrier reverse convertibles — because protection simply could not be built.
- Rates normalised and the family came back. In Germany, capital protection rose from 6.6% of investment products in March 2025 to 9.2% — €9.5bn — by June 2026.
- **But buildability is not the same as demand.**
- Swiss capital protection turnover peaked around 16% of the market in early 2023 and has since fallen back to 8%, because yield enhancement outcompetes it whenever equity volatility is well bid.

Budget figures are illustrative present-value arithmetic on annual compounding, not market quotes. The 4.3% column is the US five-year constant-maturity yield in August 2026.

WHAT ACTUALLY TRADES

Read a participation rate only against its own underlying

Tenor	Underlying	Participation	Cap
5y	S&P 500 futures index	1.40×	uncapped
5y	EURO STOXX 50	1.31×	uncapped
5y	S&P 500 futures index	1.60×	uncapped, plus a 16.5% snowball
5y	S&P 500	—	capped at 41.5%
3y	custom multi-asset index	3.54×	uncapped
5y	custom multi-asset index	6.00×	uncapped

650%

The highest advertised participation I found on a fully protected note. On a five-year at-the-money call it would imply an implied volatility of roughly 1–2%. No real equity index moves that little.

On a plain equity index with full protection you get roughly 1.0× to 1.6×, or about 8% a year capped. Every headline multiple above that is on a low-volatility custom index.

A 650% participation and a 0.95× participation can be economically identical products. The multiple tells you about the index, not about the deal.

Twin-Win: paid for being wrong, within limits

100% protected, lower barrier 80%, upper barrier 124%, participation 100% both ways



Between the barriers

The client is paid the absolute performance. Down 15% pays 15%; up 15% also pays 15%. That is the twin win.

Through either barrier

The mirror is switched off and the note redeems at the protected amount plus the minimum coupon — here, exactly 100.

Why it sits in capital protection

Because the floor is unconditional. Breaching a barrier costs the client their upside, never their principal. The family is decided by the floor, not by the barriers.

Barriers observed at the daily closing level. Note there are two of them and either one ends the mechanism — which is why a Twin-Win is much less likely to pay its headline than it first appears.

Four capital protection codes, in rising order of moving parts

1100

with Participation

A floor plus a call. The plain version, and the one the four levers describe.

1130

with Barrier

A floor plus a knocked-out call. The client participates up to a barrier; touch it and they get the floor plus a rebate. Cheaper upside, conditional.

1135

with Twin-Win

A floor plus absolute performance between two barriers, as on the previous slide.

1140

with Coupon

A floor plus a coupon and no equity option at all. At its simplest this is an accrual note.

The accrual note, and why it is here

Two-way early termination — a Holder Put and an Issuer Call on the same dates — is rare in this product set. Almost everything else gives the option to one side only.

A one-month note accruing 3.3% a year on a 30/360 basis, redeemable by the holder on any business day and callable by the issuer on any business day. There is no option on anything. It is a money-market instrument wearing a structured-product wrapper, and it is the cleanest illustration in the series that a category code describes a payoff shape, not a level of complexity.

What 'capital protected' does not mean

It applies at maturity only

Between issue and maturity the note trades at bond floor plus option value — and it will trade below par when rates rise. Duration risk is real, and selling early can crystallise a substantial loss despite the protection.

It is issuer credit, not a guarantee

No collateral, no segregated pool, no deposit insurance. The client is an unsecured senior creditor and nothing more.

The benchmark is a bond, not cash

At 4.3% over five years the issuer's own senior bond returns about 24% with certainty. The protected note returns zero if the underlying is flat or down. The client has swapped a certain 24% for a call option on 19 points of budget.

Fees come out of participation

The fee is taken from the option budget, not from the floor — so it never reduces the protection, it silently reduces the upside. At the top of the published range for total costs, a five-year note can lose roughly half its participation to fees.

Lehman's 100% Principal Protection Notes

6 cents

on the dollar

Where the notes traded after Lehman filed on 15 September 2008

\$2.5m

FINRA fine

Against the distributor, 11 April 2011, plus \$8.25m of restitution

100%

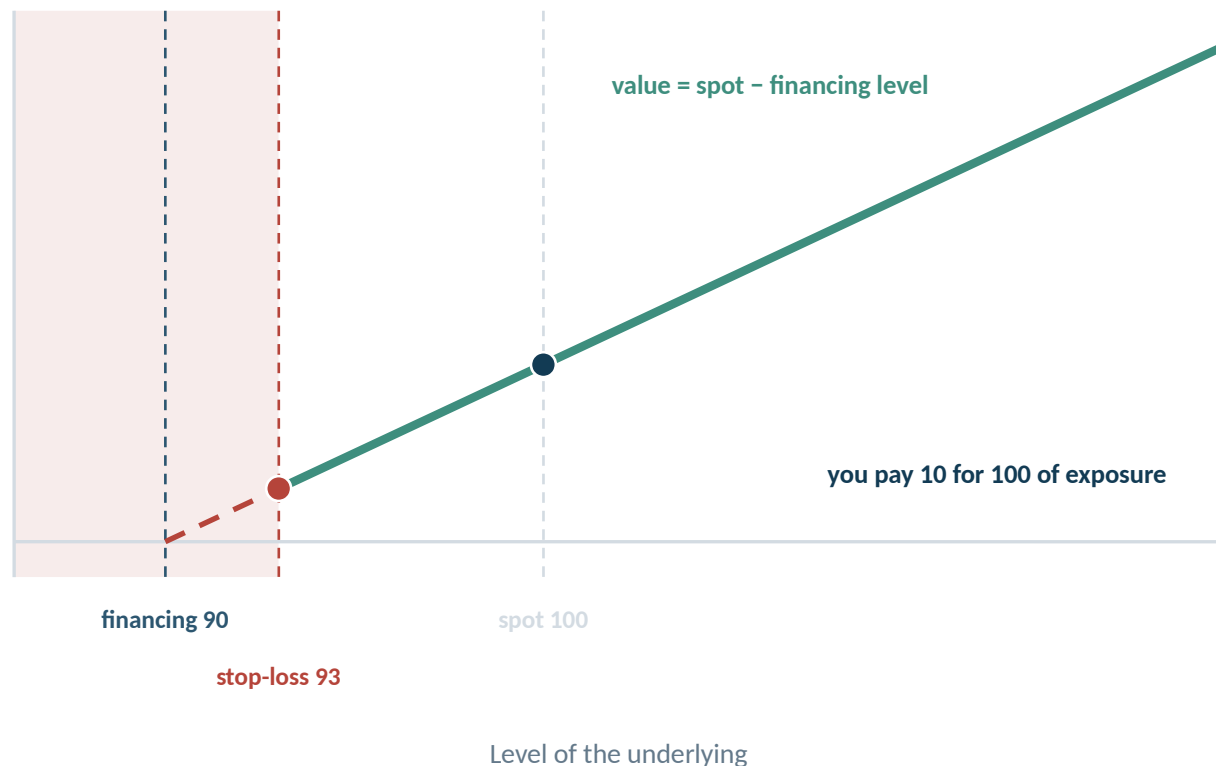
the protection level

Which was accurate, and completely beside the point

- **The notes did exactly what the term sheet said. The term sheet was not the problem.**
- The regulator's findings were that marketing suggested the protection was a guarantee while downplaying that Lehman's own credit could cause a total loss, and that the firm failed to train its advisers on Lehman's widening credit spreads.
- The regulator's summary of the sales force was blunt: they did not understand the products they were selling.
- **A separate lesson worth keeping straight: Hong Kong's Lehman 'minibonds' were credit-linked notes, not protected notes.**
- They failed through reference-entity and collateral risk, not a broken guarantee. Sixteen distributing banks repurchased them at 60% of nominal in 2009 — 70% for clients aged 65 and over — and at up to 96.5% after a collateral recovery agreement in 2011.

Three levels, and the gap between two of them is the product

Long mini future — underlying at 100, financing level 90, stop-loss 93



Financing level

The part of the underlying the issuer finances. The client pays only the gap — 10 here — and controls 100 of exposure.

Stop-loss

Set above the financing level for a long. Touch it and the product terminates immediately. It is not the same as the financing level, and that difference is the whole design.

Leverage

$\text{Spot} \div (\text{spot} - \text{financing level}) = 10\times$. It is not fixed: it rises as the position moves against you and falls as it moves for you.

A 5% rise in the underlying is a 50% gain on the certificate. A 7% fall terminates it. Both statements come from the same 10× leverage.

The carry is charged through the strike, not as a fee

- The financing level is reset every day and grows by the reference rate plus the issuer's financing spread — typically two to three percent a year.
- For a long, that means the financing level creeps upward. The stop-loss is dragged up behind it, so the buffer between spot and the barrier erodes even in a completely flat market.
- There is no line item for this. It is not a management fee, it is not a spread on the quote. It is the strike moving.
- **For a short, the signs reverse.**
- The client receives the reference rate less the spread, and the financing level falls — so a short mini future can carry positively when rates are above the spread. Very few retail clients know this.
- Dividends are deducted from the financing level on the ex-date, adjusted for withholding.

Long

Financing level rises daily.
Stop-loss follows it up.
Buffer shrinks.
Time works against you.

Short

Financing level falls daily.
Stop-loss follows it down.
Buffer grows.
Time can work for you.

A stop-loss pays something. A knock-out usually does not.

Mini future

The stop-loss sits above the financing level, so at the moment it triggers the product still has intrinsic value. The issuer unwinds its hedge and pays out what it recovers.

Residual value = (reference price – financing level) ÷ ratio

Knock-out warrant

The barrier and the strike are the same level. So when the barrier is hit, intrinsic value is zero by construction and the product expires worthless.

There is no buffer, so there is nothing to recover.

- **The reference price is not the barrier level, and it is not the price at the moment of the touch.**
- It is the worst price reached during a defined unwind window after the event — commonly the next three trading hours, extending into the following session if the market closes first.
- So the client is still exposed to further adverse movement after their position has already terminated. Determination is fast, but it is not instantaneous.
- **And it can still be a total loss.**
- If the market gaps straight through to the financing level or beyond, the residual is zero — settled at a nominal fraction of a cent. The buffer protects against an orderly move, not a gap.

WORKED EXAMPLE

Ten times leverage, in both directions

Long mini future • underlying 100 • financing level 90 • stop-loss 93 • ratio 1:1 • certificate price 10 • leverage 10×

Underlying rises 5% to 105	15.00	+50%	Value is 105 – 90. A 5% move became a 50% gain — that is the leverage working for you.
Underlying falls 3% to 97	7.00	–30%	Still alive, but the buffer to the stop-loss is now four points rather than seven, and the leverage has risen to 24×
Underlying falls 7% to 93	≈ 2.00	≈ –80%	Stop-loss touched. Terminated. If the worst price over the unwind window is 92, the residual is 2 — not the 3 the barrier implied.
Underlying gaps to 88 overnight	0.00	–100%	Straight through the financing level. Nothing to recover, and no margin call to warn anyone. Total loss.

A live Swiss single-stock mini future currently offers 11× leverage on a 7.6% stop-loss buffer.

TELLING THEM APART

Mini future against everything it gets confused with

	Mini future	Knock-out warrant	Plain warrant	CFD	Listed future
On a barrier event	Terminates, residual paid	Terminates, usually worthless	No barrier	—	—
Barrier vs strike	Two levels; the gap is the buffer	Same level	—	—	—
Volatility sensitivity	Essentially none	Low	High	None	None
Time decay	None; carry via the strike	Minimal	Yes, to zero	Daily swap charge	In the basis
Margin calls	No	No	No	Yes	Yes
Tenor	Open-ended	Dated or open-ended	Dated	Open-ended	Dated, roll needed
Counterparty	Issuer credit	Issuer credit	Issuer credit	Broker credit	Clearing house

A mini future is a security you buy outright; a CFD is a margined contract with your broker.

Tiny stock, enormous flow, and a regulator's attention

74.2%

of retail investors lost money

German study of 543,000 investors and 113m transactions, 2019–2023

€6,358

average loss

Per losing customer, over €3.4bn in total

4% / 86%

of volume vs of turnover

A sliver of the German market, most of its exchange activity

6.95

average risk indicator

Against 2.58 for the German market as a whole, on a 1–7 scale

Germany, from 16 June 2026

A mandatory warning stating that seven in ten retail investors lose money; a knowledge test requiring six correct answers out of eight, repeated every six months; and a ban on incentives including waived order fees. No leverage cap.

Netherlands, from October 2021

Leverage caps by asset class — 30× on major currencies down to 5× on shares and 2× on crypto — plus a firm-specific loss warning and an incentive ban. No knowledge test.

Regulators originally left securitised leverage outside the CFD rules because it has no margin call and a capped downside. The loss data turned out to be almost identical, so they applied CFD-style remedies anyway. Expect further convergence.

WHAT TO TAKE AWAY

Five things about protection and leverage

- 1 Participation is a residual, not a feature. Budget divided by the cost of the option, nothing more.
- 2 At zero rates a protected note cannot be built at all. The rate environment decides what is possible.
- 3 'Capital protected' means at maturity, by this issuer. It is a promise, priced on credit.
- 4 A mini future's buffer is the gap between the stop-loss and the financing level — and carry eats it daily.
- 5 Both ends of this deck are unsecured claims on the same balance sheet. That is the only thing they share.

End of the series. Six decks, four product families, one recurring question: what has the client sold?